Highlights

Conditional Inductive Guardrails: Identifying When Physical Priors Enhance, Mimic, or Impair Decisions in Wind Power Forecasting

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- Physical priors act as conditional guardrails rather than universal enhancers.
- In sample restriction, authentic priors cut risk; scrambled priors add +61.2%.
- At large samples, unconstrained deep models match or surpass fixed priors.
- Weak penalties exhibit non-specific numerical mimicry without physical bias.
- Point metrics and standalone cashflows decouple from true operational risk.

Conditional Inductive Guardrails: Identifying When Physical Priors Enhance, Mimic, or Impair Decisions in Wind Power Forecasting

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Abstract

Physical constraints are widely credited with improving wind power forecasting, but error reductions often conflate physical guidance with capacity shifts, auxiliary data budgets, non-specific regularization, proxy misalignment, or metric artifacts. Across three commercial UK wind farms holding backbone, capacity, data splits, and calibration constant, we contrast authentic empirical power curve priors against form- and weight-matched pseudo-scrambled negative controls across sample regimes, penalty scales, and proxy latencies across five synchronized random seeds. In sample-restricted regimes (N_{train} = 1,583, Penmanshiel), strong empirical curve constraints (λ = 1.0) reduce asymmetric dispatch loss by 1.28%, whereas permuting the aerodynamic mapping triggers a 61.2% surge in dispatch loss, providing evidence of prior-specific mapping effects in this tested setting. A budget-matched information control ($N_{\text{prior}} = N_{\text{train}} = 500$) confirms that domain-structured priors retain 77%–92% of the full-history benefit, indicating that most observed gains cannot be explained by additional data volume alone. However, as training data expand to N_{train} = 5,000, unconstrained data-driven models match or slightly outperform fixed priors under the tested strong-prior setting (+0.33% on Kelmarsh, +1.73% on Hill of Towie). Under weak regularization (λ = 0.01), genuine aerodynamic curves and scrambled negative controls yield closely matched outcomes, so observed gains cannot be specifically attributed to physical mapping and remain consistent with non-specific numerical regularization. Training objectives regularized against lagged weather proxies exceed the unconstrained reference by the first evaluated lag (k = 6, 1 h) on Hill of Towie and deteriorate monotonically through k = 36, while continuous corruption crosses the baseline around $\alpha \approx 0.07$ on Penmanshiel. Downstream, symmetric point accuracy (RMSE) decouples from operational dispatch loss, and a degenerate zero-bid baseline mechanically generates six-figure balancing cashflows without forecasting skill. Auditing 46 contemporary studies reveals that surveyed publications reported exclusively within nominal envelopes, while 50.0% relied on confounded module-level ablations. Physical priors should therefore be evaluated not as universal enhancers, but as conditional inductive guardrails requiring negative controls, boundary stress tests, and decision-aligned evaluation.

Keywords: Physics-Constrained Deep Learning, Wind Power Forecasting, Prediction-Decision Decoupling, Conditional Inductive Guardrails, Attributional Identification, Reproducibility and Auditing, Machine Learning in Energy Systems

1. Introduction

In electric power systems and wholesale electricity markets, the operational value of wind power forecasting is governed by asymmetric decision costs arising from reserve allocation, economic dispatch commitments, and real-time balancing settlement [1, 2]. Under standard market clearing and system balancing rules, reserve shortfall penalties for under-delivery are typically substantially more severe than downward redispatch or spill penalties for over-delivery [3]. In these asymmetric operational regimes, conventional symmetric regression metrics, such as Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE), fail to reflect tail operational risk and downside financial exposure [4, 5]. To improve generalization and enforce physically realistic outputs, an expanding body of literature embeds aerodynamic limits, empirical power curves, kinematic wake equations, and joint probability distribution constraints

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directly into deep neural network architectures and loss functions [6, 7, 8, 9, 10]. A foundational premise underpins this research paradigm: that reported reductions in forecasting error can be directly attributed to the valid physical domain knowledge encoded within the constraint.

However, observing that a physics-constrained model outperforms an unconstrained baseline does not establish that the gain arises from genuine domain knowledge. In empirical machine learning, performance differences between candidate models can be driven by multiple competing mechanisms:

(a) *Parametric capacity shifts*: modifying neural submodules or layer architectures when adding constraints confounds physical bias with representational capacity;

(b) *Information-budget leakage*: fitting empirical reference priors on historical datasets provides extra observation volume unavailable to the base forecaster;

(c) *Non-specific numerical regularization*: additional penalty objectives can act as generic smoothers or L2 regularizers rather than imparting domain aerodynamic understanding;

(d) *Structural approximation error*: fixed algebraic equations or static reference curves inevitably introduce approximation bias as data availability grows;

(e) *Temporal proxy misalignment*: regularizing offline training targets against lagged weather telemetry introduces state distortion when forward meteorological forecasts are absent;

(f) *Decision-metric misalignment*: improvements under symmetric point metrics frequently fail to translate into downside risk reductions under asymmetric market dispatch.

Because conventional benchmarking protocols rarely isolate these confounding vectors, existing literature leaves the foundational attribution question unresolved: When does a physical prior provide genuine decision value, when is its apparent benefit non-specific, and when does it become harmful?

In this investigation, we address this question by evaluating physical priors as **conditional inductive guardrails** rather than universally beneficial performance enhancers. The central scientific problem is to identify when a physical prior provides genuine decision value, when its apparent benefit is non-specific, and when it becomes harmful. To delineate this operational envelope, we introduce a **Controlled Attribution Protocol** across three commercial UK wind farms (Kelmarsh, Hill of Towie, and Penmanshiel) coupled with 2017 UK Elexon Balancing and Settlement Code (BSC) market clearing prices. Holding forecasting backbones, neural capacity, data splits, and post-hoc quantile calibration strictly constant, we contrast authentic empirical power curve priors against form- and weight-matched pseudo-scrambled negative controls across varied sample regimes, constraint weights, and proxy delays.

Our controlled experiments establish that physical priors exhibit conditional, context-dependent operational utility:

1. **Prior Specificity under Sample Restriction**: In the tested sample-restricted regime on Penmanshiel ($N_{\text{train}} = 1{,}583$), strong empirical curve constraints ($\lambda = 1.0$) reduce normalized asymmetric dispatch loss by 1.28%, whereas permuting the aerodynamic mapping via a pseudo-scrambled control inflates dispatch loss by 61.2%, providing empirical evidence of prior-specific mapping effects in this sample-restricted setting.

2. **Diminishing Advantages across Sample Budgets**: Across within-site learning curves ($N_{\text{train}} \in [500, 5000]$), fixed empirical priors provide cost reductions in the tested restricted windows ($N_{\text{train}} \in \{500, 1{,}000\}$). A strictly budget-matched information control ($N_{\text{prior}} = N_{\text{train}} = 500$) confirms that domain-structured prior effects retain 77% to 92% of the full-history benefit, demonstrating that the majority of the observed benefit cannot be explained by additional historical data volume alone. However, at the largest evaluated training budget ($N_{\text{train}} = 5{,}000$), unconstrained data-driven models match or slightly outperform fixed priors (+0.33% on Kelmarsh, +1.73% on Hill of Towie), consistent with a transition where the regularizing value of a fixed prior diminishes while structural approximation error becomes increasingly consequential.

3. **Non-Specific Numerical Regularization Mimicry**: Under weak penalty scaling ($\lambda = 0.01$), authentic and scrambled priors yield closely matched outcomes (0.08803 vs. 0.08792 on Kelmarsh; 0.08913 vs. 0.08910 on Hill of Towie), indicating that the observed effect cannot be specifically attributed to physical mapping and remains consistent with non-specific numerical regularization.

4. **Empirical Boundary Crossovers**: Continuous stress testing on Penmanshiel reveals that the evaluated cor-

ruption trajectory crosses the unconstrained baseline around an empirical crossover region of $\alpha \approx 0.07$, while turbine capacity derating exceeding 10% flips physical priors into net economic penalties under the tested configuration. In the Hill of Towie lag scan, decision risk exceeds the unconstrained reference by the first evaluated lag ($k = 6$ / 1 h delay) and deteriorates monotonically through $k = 36$.

5. **Prediction–Decision Decoupling**: In an info-symmetric evaluation on Kelmarsh, adding wind information improves RMSE by 1.22% yet worsens asymmetric decision cost by 1.35%. Downstream, a degenerate zero-bid baseline ($q \equiv 0$) mechanically generates six-figure apparent balancing cashflows without forecasting skill, exposing standalone unhedged balancing cashflows as an unreliable proxy for operational performance.

To contextualize these empirical failure regimes within current reporting practices, we conduct an evidence-gap literature audit of 46 recent studies (2023–2026) and a supporting static computational graph audit of the sole open-source end-to-end training codebase in this domain. Studies in this scoping cohort reported exclusively within nominal operating envelopes (0/46 documenting operational failure regimes), while 44/46 lacked public code and 50.0% relied exclusively on confounded module ablations. Concurrently, static AST inspection of the Physics-Constrained Transformer (PCT) uncovers an inverted argument order in its divergence loss, producing an analytical $O(1/\epsilon)$ gradient response under small predicted probabilities while its bivariate kernel coupling remains mathematically sound.

1.1. Summary of Contributions

This investigation establishes an evidence-calibrated foundation for physics-constrained wind power forecasting through three interconnected contributions:

- **Controlled Attribution Protocol for Physical Priors**: We formulate and execute a controlled benchmarking framework that pairs authentic physical priors with form- and weight-matched pseudo-scrambled negative controls under identical network architectures, optimization schedules, and post-hoc quantile calibrations, providing an empirical standard to distinguish domain aerodynamic guidance from non-specific numerical smoothing.

- **Empirical Operating Envelope Mapping**: Across three commercial wind farms, we map the operational boundaries of empirical power-curve priors: documenting the attenuation of prior advantages as sample size increases, confirming via 500-sample budget matching that benefits cannot be explained by historical volume alone, and characterizing continuous prior corruption ($\alpha \approx 0.07$ crossover on Penmanshiel) and training proxy staleness (where decision risk exceeds the unconstrained reference by the first evaluated lag $k = 6$ / 1 h delay on Hill of Towie and deteriorates monotonically through $k = 36$).

- **Decision-Aligned Evaluation and Governance Framework**: Integrating a decision-theoretic risk decomposition, formal counterexamples, and empirical evaluations (info-symmetric decoupling and zero-bid reductio), we demonstrate why symmetric point metrics and standalone balancing cashflows fail to reflect asymmetric dispatch costs. We translate these findings into a five-step practical decision rule and a five-dimension methodological reporting checklist (C1–C5) for physics-constrained forecasting.

Empirical Scope and Methodological Boundaries. To prevent over-generalization, we explicitly delineate four empirical boundaries governing this study: first, our literature audit characterizes reporting patterns within the 46 studies identified under our prespecified PRISMA-ScR protocol (2023–2026), rather than asserting universal claims across all scientific machine learning; second, software auditing is focused on the computational graph of the open-source PCT benchmark; third, prospective empirical ablations evaluate boundary behaviors across three commercial UK wind farms under matched DLinear backbones, with PatchTST serving as an architectural sensitivity probe; and fourth, downstream evaluations analyze operational decision risk and stylized imbalance cashflow component diagnostics under UK BSC prices, characterizing metric decoupling rather than full commercial portfolio trading.

2. Related Work and Evidence-Gap Literature Audit

To contextualize why existing literature leaves the attribution question unresolved, we examine three intersecting research domains: (i) physics-constrained formulations in renewable energy forecasting, (ii) operational decision evaluation in electricity markets, and (iii) reproducibility and auditing standards in scientific machine learning. We then present a concise evidence-gap scoping audit across 46 contemporary studies, demonstrating that publications in the surveyed scoping cohort systematically omitted the negative controls and boundary stress tests required to establish domain-specific attribution.

2.1. Physical Constraints in Wind Power Forecasting

Embedding physical domain knowledge into deep neural networks aims to combine governing physical principles with data-driven function approximation [7, 8]. In wind power forecasting, physical constraints typically follow three main formulations:

Operational Envelope Bounds. Enforcing aerodynamic operating thresholds (such as cut-in, rated, and cut-out wind speeds, or Betz envelope limits) via bounded activations or post-hoc clipping guarantees that outputs respect nameplate boundaries [11]. However, hard saturation can alter gradient flows near boundary cutoffs. In turbulent boundary-layer conditions where turbine generation hovers near rated capacity, zero-gradient saturation regions can impede gradient descent, restricting local adaptation to rapid meteorological transitions.

Kinematic Wake Formulations. Spatio-temporal forecasters commonly embed steady-state kinematic wake models—such as the 1D Jensen or Bastankhah–Porté-Agel Gaussian deficit equations [12, 6, 9]—into Graph Neural Network (GNN) adjacency matrices. While this incorporates spatial layout geometry, replacing dynamic Navier–Stokes fluid mechanics with static engineering equations can restrict network flexibility under complex atmospheric stratification. Furthermore, published studies frequently attribute empirical gains to the embedded wake model without isolating whether the expressive capacity of the GNN backbone alone accounts for the improvement.

Distributional Regularization. Penalizing statistical divergence between predicted outputs and historical joint wind speed–power distributions (via Wasserstein distance or Jensen–Shannon divergence [10, 11]) introduces complex optimization dynamics [13]. Static distribution alignment regularizes predictions toward pre-computed empirical manifolds. When meteorological regimes shift or input observations suffer from temporal latency, divergence penalties can penalize genuine operational transients, creating optimization trade-offs between distribution fidelity and immediate dispatch accuracy.

2.2. Decision-Theoretic Evaluation and Settlement Mechanics

Wind power forecasting models have traditionally been evaluated using symmetric statistical error metrics, such as MAE, RMSE, and R^2 [14, 3]. Madsen et al. [14] established standardized verification protocols highlighting the importance of evaluating models across horizons, operational regimes, and normalized capacity metrics.

However, power system operations demonstrate that statistical error reductions do not guarantee economic value under asymmetric market penalties [2, 3]. Pinson et al. [5] demonstrated that optimal bidding under asymmetric imbalance penalties is governed by conditional quantiles matching the operational cost ratio, highlighting the structural decoupling between point prediction accuracy and decision risk. Downstream grid dispatch penalizes reserve shortfalls far more severely than excess generation [15, 16]. Furthermore, real-world balancing arrangements—such as the UK Elexon Balancing and Settlement Code (BSC)—aggregate energy deviations over discrete 30-minute settlement periods. Under isolated single-component accounting, unhedged cashflows can create apparent monetary surpluses through systematic under-commitment, decoupling cashflow from forecasting skill. Evaluating forecasters strictly on uncalibrated point errors or unhedged cashflows obscures their true operational efficacy.

2.3. Reproducibility Standards and Software Auditing in Scientific AI

Empirical machine learning has faced increasing scrutiny over data leakage, underspecification, and implementation discrepancies [17, 18, 19, 4]. Kapoor and Narayanan [19] documented how subtle data leakage vectors undermine reproducibility across seventeen scientific domains. Gundersen and Kjensmo [18] evaluated 400 papers from major AI conferences, finding that fewer than 20% shared executable source code or reproducible experimental protocols.

To address these vulnerabilities, procedural frameworks have gained traction. Pineau et al. [20] developed the NeurIPS Machine Learning Reproducibility Checklist, demonstrating that explicit reporting mandates improve disclosure of code, data, hyperparameter choices, and error bounds. D'Amour et al. [17] showed that models with equivalent in-distribution test accuracy often diverge under distribution shift due to underspecified training criteria. Concurrently, Lipton and Steinhardt [4] analyzed recurring pitfalls in empirical scholarship: confusing empirical association with theoretical explanation, failing to isolate independent sources of performance gains, and employing decorative mathematics that does not govern empirical outcomes.

2.4. Evidence-Gap Scoping Audit: Why Current Evidence Cannot Settle Attribution

While general machine learning venues have increasingly adopted rigorous reporting norms, physics-

Table 1: Positioning this Scoping Empirical Audit Relative to Existing Literature Reviews in Wind Power Forecasting and Scientific ML.

Survey / Meta-Review **Literature Reference**	**Decouples** **Provenance** ($\mathcal{D}_1$)	**Audits Stage** **Separation** ($\mathcal{D}_2$)	**Checks Inference** **Telemetry** ($\mathcal{D}_3$)	**Verifies Code** **Parity** ($\mathcal{D}_4$)	**Scrutinizes** **Validity/Harm** ($\mathcal{D}_5$)
Yan et al. (2015) [21]	Partial	No	Partial	No	No
Maldonado-Correa et al. (2021) [22]	No	No	No	No	No
Karniadakis et al. (2021) [7]	Conceptual	Conceptual	No	No	No
Zhang et al. (2014) [3]	No	No	Partial	No	No
Wang et al. (2021) [23]	No	Partial	No	No	No
This Scoping Empirical Audit	**Yes (Schema v4)**	**Yes (Schema v4)**	**Yes (Schema v4)**	**Yes (Static Audit)**	**Yes (Strict Isolation)**

constrained wind forecasting has developed largely without standardized negative controls. To systematically examine current reporting patterns across the field, we conducted a scoping audit of $N = 46$ contemporary studies (2023–2026) following PRISMA-ScR guidelines (full retrieval protocols, eligibility criteria, and multi-rater reliability tests are detailed in Appendix B and Appendix C).

The scoping audit uncovers three systemic evaluation gaps:

1. **Absence of Operational Failure Reporting (0/46 studies, 0.0%)**: Across the 46 audited studies, not a single publication documented an operating regime, data budget, or stress-test condition in which physical constraints degraded downstream performance (95% Wilson CI: [0.0%, 7.7%]). Only three studies (6.5%) examined sensitivity to prior parameter variations; the remaining 43 tested constraints exclusively under nominal design configurations. Studies in the scoping cohort reported exclusively within nominal operating envelopes, leaving behavior outside nominal design conditions unexamined in the surveyed literature.

2. **Severe Open-Code Deficit (44/46 studies, 95.7%)**: Only two of the 46 studies provided executable source code or reproducible computational scripts (95% Wilson CI: [85.5%, 98.8%]). The remaining 44 studies omitted public repositories, preventing independent verification of mathematical loss implementations and computational graph executions.

3. **Confounded Module-Level Ablations (23/46 studies, 50.0%)**: Exactly half of the audited studies evaluated physical constraints by excising entire multi-parameter neural blocks (e.g., stripping GNN layers or dual-stream attention branches) rather than toggling the constraint penalty multiplier ($\lambda = 0$ vs. $\lambda > 0$) under matched network capacity (95% Wilson CI: [36.1%, 63.9%]). Because excising submodules alters parameter counts by thousands of weights, these ablations conflate representation capacity with physical inductive guidance.

As summarized in Table 1, existing literature surveys have focused predominantly on model taxonomies without auditing lifecycle provenance, software fidelity, or ablation confounding. These three structural gaps demonstrate why conventional literature comparisons cannot answer the central attribution question: without form-matched negative controls, disclosed data budgets, and unconfounded parameter ablations, observed empirical gains cannot be reliably attributed to valid physical domain knowledge.

3. Mathematical Formulation and Decision-Theoretic Foundation

In this section, we establish the mathematical foundation governing operational wind power dispatch and market settlement. We formulate the operational problem under asymmetric balancing costs, derive the exact integral representation of excess operational risk, decompose it into representation error and calibration components, and establish formal boundary counterexamples delineating the limits of scalar calibration.

3.1. Operational Dispatch Problem Formulation

Let $X \in \mathcal{X} \subset \mathbb{R}^{L \times D}$ denote historical multivariate telemetry, and let $Y \in [0, 1]$ denote normalized turbine active power output over the target delivery horizon. In operational power system balancing, transmission system operators evaluate declared bids under asymmetric

pinball dispatch penalties [2, 3]:

$$\ell(p, Y; r) = c_+ \max(p - Y, 0) + c_- \max(Y - p, 0), \quad (1)$$

where $p \in [0, 1]$ is the declared dispatch commitment, c_+ is the shortfall unit penalty (under-generation where declared power exceeds physical delivery, $p > Y$, requiring expensive tertiary reserve activation), c_- is the surplus unit penalty (over-generation, $Y > p$, inducing downward redispatch or curtailment), and $r = c_+/c_-$ represents the cost asymmetry ratio. Normalizing by c_-, the asymmetric operational loss is:

$$\mathcal{L}_{\text{disp}}(p, Y) = r \max(p - Y, 0) + \max(Y - p, 0). \quad (2)$$

The expected operational risk under policy $p(X)$ is $\mathcal{R}(p) = \mathbb{E}_{X,Y}[\mathcal{L}_{\text{disp}}(p(X), Y)]$.

3.2. Lemma 1: Exact Integral Excess Operational Risk

Lemma 1 formalizes a classical excess risk analytical identity in statistical decision theory and quantile regression, grounded in the foundational formulations of Koenker [24] and electricity market bidding principles established by Pinson et al. [5]. The unconstrained Bayes-optimal dispatch decision minimizing $\mathcal{R}(p)$ is the conditional quantile:

$$q_\tau(X) = F^{-1}_{Y|X}(\tau), \quad \text{where } \tau = \frac{1}{r+1}. \quad (3)$$

For an operational cost asymmetry ratio $r = 2$, the critical quantile is $\tau = 1/3 \approx 0.333$. Grounded in classical decision theory and proper scoring rules [24, 15, 5], we express the relationship between point forecast deviation and downstream dispatch cost through the following integral identity:

Lemma 1 (Decision-Theoretic Excess Operational Risk [24, 15, 5])**.** *For any dispatch decision rule $p(X) \in [0, 1]$, the excess operational risk relative to the Bayes-optimal quantile $q_\tau(X)$ satisfies the exact integral identity:*

$$\mathcal{R}(p) - \mathcal{R}(q_\tau) = (r+1)\,\mathbb{E}_X\left[\int_{q_\tau(X)}^{p(X)} \Big(F_{Y|X}(u) - \tau\Big)\,du\right] \geq 0, \quad (4)$$

which is non-negative everywhere. Under standard regularity conditions where $F_{Y|X}$ is strictly increasing on its support such that $q_\tau(X)$ is uniquely defined, equality holds if and only if $p(X) = q_\tau(X)$ almost surely.

Proof. For a fixed feature vector X, the expected conditional loss at commitment p is:

$$\mathbb{E}[\mathcal{L}_{\text{disp}}(p, Y) \mid X] = r \int_0^p (p - y)dF(y) + \int_p^1 (y - p)dF(y), \quad (5)$$

where $F(y) = F_{Y|X}(y)$. Taking the partial derivative with respect to p:

$$\begin{aligned} \frac{\partial}{\partial p}\mathbb{E}[\mathcal{L}_{\text{disp}}(p, Y) \mid X] &= rF(p) - (1 - F(p)) \\ &= (r+1)F(p) - 1. \end{aligned} \quad (6)$$

Recalling that $\tau = 1/(r+1)$, this gradient simplifies to:

$$\frac{\partial}{\partial p}\mathbb{E}[\mathcal{L}_{\text{disp}}(p, Y) \mid X] = (r+1)\Big(F(p) - \tau\Big). \quad (7)$$

By the Fundamental Theorem of Calculus, integrating the gradient from $q_\tau(X)$ to $p(X)$ yields:

$$\begin{aligned} &\mathbb{E}[\mathcal{L}_{\text{disp}}(p, Y) \mid X] - \mathbb{E}[\mathcal{L}_{\text{disp}}(q_\tau, Y) \mid X] \\ &\quad = (r+1)\int_{q_\tau(X)}^{p(X)} \Big(F(u) - \tau\Big)du. \end{aligned} \quad (8)$$

Because $F(u)$ is monotonically non-decreasing, for any $u > q_\tau(X)$ we have $F(u) \geq \tau$, so the integrand is non-negative; for any $u < q_\tau(X)$ we have $F(u) \leq \tau$, and integrating backwards from $q_\tau(X)$ to $p(X)$ also yields a positive integral. Taking the outer expectation over $\mathcal{X}$ completes the proof. □

3.3. Two-Stage Decomposition and Decoupled Decision Calibration

When $p(X)$ is in the neighborhood of $q_\tau(X)$, expanding $F_{Y|X}(u)$ via a first-order Taylor series around $q_\tau(X)$ yields:

$$F_{Y|X}(u) - \tau \approx f_{Y|X}(q_\tau(X)) \cdot (u - q_\tau(X)), \quad (9)$$

which produces the local quadratic excess risk approximation:

$$\mathcal{R}(p) - \mathcal{R}(q_\tau) \approx \frac{r+1}{2}\,\mathbb{E}_X[f_{Y|X}(q_\tau(X)) \cdot (p(X) - q_\tau(X))^2]. \quad (10)$$

Now consider the standard two-stage paradigm where a forecaster predicts $f_\theta(X)$ aiming for conditional expectation $\mu(X) = \mathbb{E}[Y \mid X]$, and the dispatch commitment is formed via a scalar quantile offset δ: $p(X) = f_\theta(X) + \delta$. Expressing the prediction error as:

$$p(X) - q_\tau(X) = \big(f_\theta(X) - \mu(X)\big) + \big(\delta - \Delta_\tau(X)\big), \quad (11)$$

where $\Delta_\tau(X) = q_\tau(X) - \mu(X)$ denotes the conditional quantile-mean shift. Under homoscedastic curvature $w_\tau = \frac{r+1}{2} f_{Y|X}(q_\tau)$, substituting into Eq. (10) decomposes excess operational risk into three distinct terms:

$$\begin{aligned} \mathcal{R}(p) - \mathcal{R}(q_\tau) \approx\ & w_\tau\,\mathbb{E}_X[(f_\theta(X) - \mu(X))^2] \\ & + w_\tau\,\mathbb{E}_X[(\delta - \Delta_\tau(X))^2] \\ & + 2w_\tau\,\mathbb{E}_X[(f_\theta(X) - \mu(X))(\delta - \Delta_\tau(X))], \end{aligned} \quad (12)$$

where the components represent **Term I** (the risk of the calibrated estimator under the empirical representation), **Term II** (Quantile Mismatch / Calibration Dispersion), and **Term III** (Cross-Coupling Interaction Component), respectively.

We explicitly emphasize that this three-term decomposition is decision-theoretic and mathematically distinct from the classical statistical bias–variance decomposition: Term I represents the risk of the calibrated estimator under the empirical representation, measuring the expected squared prediction error of the forecaster $f_\theta(X)$ relative to the true conditional expectation $\mu(X)$, and is never equated with prediction variance or parameter-estimation variance.

3.4. Decision-Theoretic Deductions from the Three-Term Decomposition

The three-term decomposition in Eq. (12) provides a structured lens for interpreting how point predictions and quantile offsets interact:

Remark 1 (Deduction 1: Pseudo-Advantage Masking in Uncalibrated Regimes). In uncalibrated forecasting regimes ($\delta = 0$), Term II ($w_\tau \mathbb{E}[\Delta_\tau(X)^2]$) is an irreducible data-dependent constant representing the dispersion between conditional mean and target quantile. A physics-regularized model with inferior conditional mean accuracy (larger Term I) can paradoxically achieve lower uncalibrated operational risk if its structural bias ($f_\theta(X) - \mu(X)$) aligns with the negative quantile gap $\Delta_\tau(X) = q_\tau(X) - \mu(X) < 0$ (for $r = 2, \tau = 1/3$). This alignment drives the interaction Term III ($2w_\tau \mathbb{E}[(f_\theta(X) - \mu(X))(-\Delta_\tau(X))] < 0$) negative, accidentally cancelling out a portion of representation error. Upon post-hoc quantile calibration, δ absorbs this systematic offset, neutralizing Term III and revealing that the apparent physical advantage was an artifact of uncalibrated evaluation.

Diagnostic Caveat: Term III serves as an analytical diagnostic framework for explaining how structural bias can mask forecasting error under uncalibrated evaluation, rather than an empirically estimated scalar quantity. In operational telemetry where the true conditional distribution $F_{Y|X}$ is unknown, isolating the exact point-by-point magnitude of Term III is intractable without strong parametric assumptions.

Remark 2 (Deduction 2: Regularization Distortion under Well-Sampled Regimes). Optimal quantile calibration ($\delta^* = \arg\min_\delta \hat{\mathcal{R}}(f_\theta + \delta)$) removes the unconditional component of Term II. In well-sampled regimes where unconstrained models already achieve low representation error (Term I), injecting an uncalibrated physical loss penalty can act as an uncompensated structural distortion that inflates Term I, providing an analytical mechanism consistent with the empirical degradation observed under fixed priors in well-sampled settings.

Remark 3 (Deduction 3: Inductive Guidance and Prior Sensitivity under Sample Restriction). In operating regimes with restricted effective sample size, unconstrained empirical estimation is prone to over-fitting and elevated representation error under the empirical data sample (manifested as an inflated Term I). Under sufficiently strong penalty weight ($\lambda = 1.0$), genuine physical constraints provide inductive regularizing guidance that suppresses non-physical parameter excursions, securing robust downside risk reduction. Conversely, when this mapping is misspecified or permuted (as in our scrambled-prior negative control), the rigid penalty enforces counter-physical gradient updates, precipitating severe performance degradation and demonstrating acute vulnerability to prior misspecification under strong regularization.

3.5. Formal Boundary Counterexamples (Possibility Results)

To ensure theoretical precision and delineate the limits of scalar calibration, we establish two formal possibility counterexamples demonstrating that decoupled calibration dominance is conditional rather than absolute.

Under *Constant Shift Invariance* (Counterexample 1), if a physical constraint modifies a forecaster purely by a uniform additive shift $f_{\text{phys}}(X) = f_{\text{uncons}}(X) + c$, the subsequent optimal scalar calibration absorbs c completely ($\delta^*_{\text{phys}} = \delta^*_{\text{uncons}} - c$), resulting in identical dispatch decisions and zero excess risk difference ($\Delta\mathcal{R}_{\text{disp}} \equiv 0$) despite an apparent increase in prediction MSE. Under *Spatial Heteroscedasticity* (Counterexample 2) with input-dependent variance $\sigma(X)$, a global scalar calibration fails to compensate for localized variance shifts; here, an estimator that introduces structural bias and exhibits higher MSE can achieve strictly lower operational risk than the true conditional mean following scalar quantile offset. These existence results prove that while scalar calibration successfully decouples mean prediction from global quantile alignment, it does not guarantee Pareto dominance over locally adaptive or asymmetric loss training under severe conditional heteroscedasticity. Complete mathematical proofs for both counterexamples are provided in Appendix A.

3.6. Two-Settlement Wholesale Accounting and Real-Time BSC Imbalance Netting

In wholesale electricity markets, wind power producers operate within a two-settlement framework coupling

forward contracts and physical real-time imbalance settlement [15, 5]. Total commercial revenue is expressed as:

$$R_{\text{total}} = P_{\text{F}} \cdot q + P_{\text{imb}} \cdot (y - q), \tag{13}$$

where P_{F} denotes the forward contract price (e.g., bilateral PPA or day-ahead exchange clearing price), q is the forward-contracted or nominated energy position in the stylized model, y is actual physical generation volume, and P_{imb} is the system imbalance cash-out price. Under the UK Balancing and Settlement Code (BSC) single-price regime (post-P305), energy imbalance cashflows aggregate over discrete half-hour Settlement Periods (SP):

$$E_{\text{SP}} = \sum_{t \in \text{SP}} (y_t - q_t) P_{\text{rated}} \Delta t \quad [\text{MWh}], \quad \Delta t = \frac{1}{6}\ \text{h}, \tag{14}$$

where P_{rated} denotes total wind farm nameplate capacity in MW. Physical shortfall occurs when $E_{\text{SP}} < 0$, incurring settlement charges evaluated at the System Buy Price λ_t^{SBP}. The heavy-tailed price distribution (mean £44.27/MWh, peak £1,509.80/MWh) acts as a high-magnitude risk multiplier. When extreme price spikes coincide with wind ramp errors, the cash-out penalty relative to average market conditions is amplified by up to 34.1× (reflecting the ratio of the annual peak SBP of £1,509.80/MWh to the mean SBP of £44.27/MWh). However, as demonstrated in Section 7, evaluating forecasters on isolated unhedged balancing cashflows $P_{\text{imb}} \cdot (y - q)$ without forward revenue $P_{\text{F}} \cdot q$ creates severe metric distortion, establishing that non-negative asymmetric decision loss serves as the primary ground-truth metric.

4. Controlled Empirical Testbed and Attributional Protocol

This section details the operational wind farm testbed, the five-dimensional constraint lifecycle taxonomy, and our Controlled Attribution Protocol designed to distinguish genuine domain physical guidance from confounding alternative mechanisms.

4.1. The Five-Dimensional Constraint Lifecycle Taxonomy

Designating a neural forecaster as "physics-constrained" often groups fundamentally disparate mathematical, architectural, and data-flow operations under a single heading [4]. To establish an operational foundation for systematic auditing and controlled benchmarking, we define a structured **Constraint Lifecycle Taxonomy**.

The primary unit of analysis is the individual forecasting model, characterized by its operational constraint mechanism. Formally, we characterize the primary constraint architecture C as a 5-tuple operating across orthogonal lifecycle stages:

$$C = \langle \mathcal{D}_1, \mathcal{D}_2, \mathcal{D}_3, \mathcal{D}_4, \mathcal{D}_5 \rangle, \tag{15}$$

where $\mathcal{D}_1$ represents parameter provenance, $\mathcal{D}_2$ denotes application stage, $\mathcal{D}_3$ specifies inference telemetry requirements, $\mathcal{D}_4$ captures implementation software fidelity, and $\mathcal{D}_5$ assesses evaluation validity and harm reporting (Figure 1).

Dimension 1: Parameter Provenance ($\mathcal{D}_1$). Identifies the calibration origin of constraint parameters: (1) *intrinsic*, where parameters derive analytically from physical invariants without empirical fitting (e.g., Betz limit $C_p \leq 16/27$ or kinematic wake equations); (2) *historical-static priors*, where reference power curves or joint probability distributions are estimated offline on historical SCADA archives and held static during deployment; (3) *learned-online*, where parameters adapt dynamically via auxiliary neural subnetworks; and (4) *unreported*, where calibration details are omitted.

Dimension 2: Application Stage ($\mathcal{D}_2$). Identifies where domain physics enters the computational pipeline: (1) *training-loss regularization* ($\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda \mathcal{L}_{\text{phys}}$); (2) *inference-architecture constraints*, modifying layers to project outputs onto admissible manifolds; (3) *data-generation priors*, synthesizing augmented training scenarios; and (4) *post-hoc correction filters*, applied externally after inference.

Dimension 3: Inference Telemetry Requirements ($\mathcal{D}_3$). Identifies external operational data streams required during real-time deployment: (1) *none*, relying solely on target turbine historical power; (2) *forecast-input*, requiring future numerical weather prediction (NWP) inputs; and (3) *contemporaneous-state*, requiring synchronized real-time measurements from adjacent turbines or met masts.

Dimension 4: Implementation Fidelity ($\mathcal{D}_4$). Evaluates whether public software faithfully executes published mathematical formulations: (1) *code-aligned*, where public source code strictly matches published equations; (2) *code-mismatch*, where the codebase exhibits an algebraic or directional deviation; and (3) *not-found*, where no public repository or script accompanies the study.

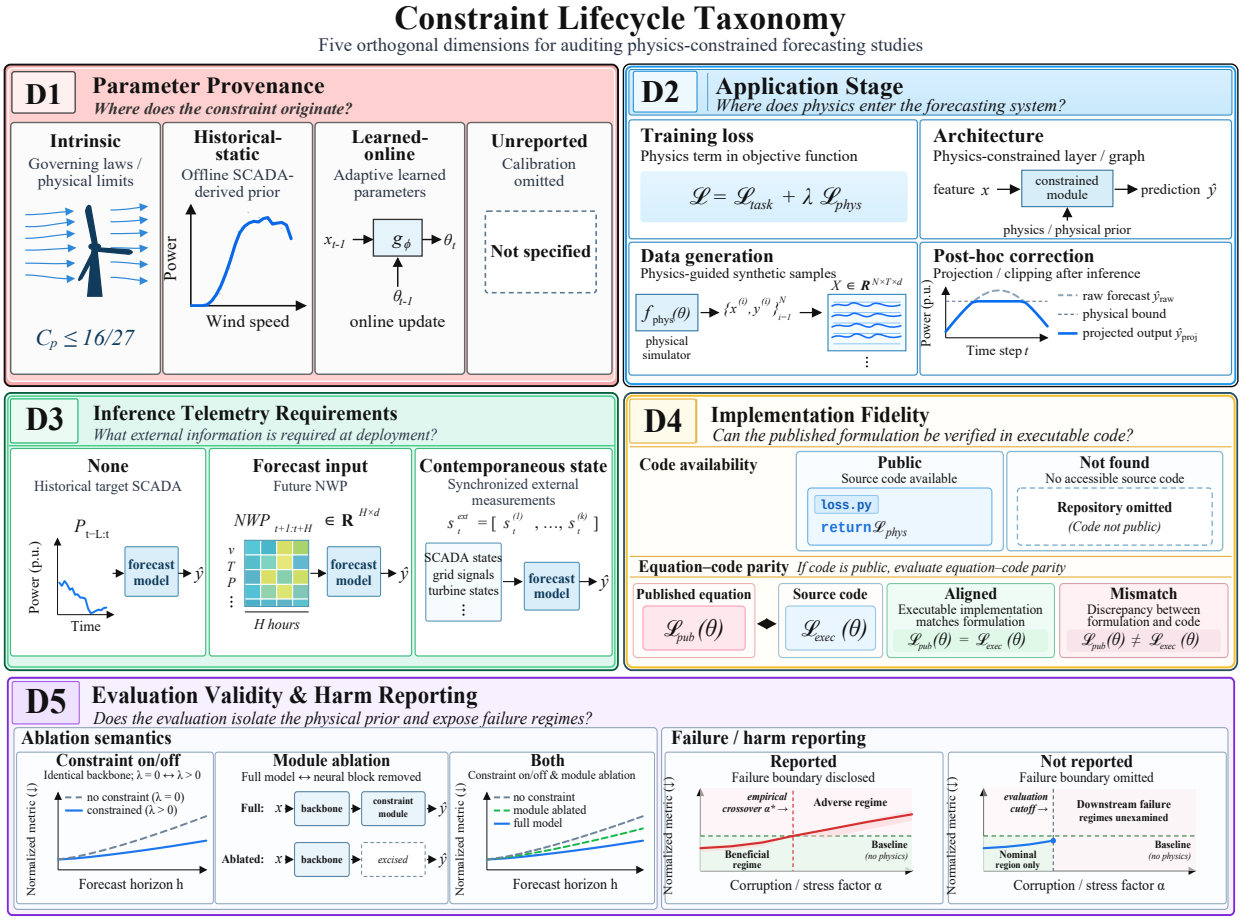


Figure 1: The Five-Dimensional Constraint Lifecycle Framework. Tracking domain physical constraints from parameter origin ($\mathcal{D}_1$) through computational graph formulation ($\mathcal{D}_2$), operational telemetry dependency ($\mathcal{D}_3$), software implementation parity ($\mathcal{D}_4$), and evaluation validity and harm reporting ($\mathcal{D}_5$).

Dimension 5: Evaluation Validity and Harm Reporting ($\mathcal{D}_5$). This dimension contains two complementary subfields: (i) *ablation semantics*, classified as *constraint-on-off* (where treatment and baseline share identical parameter capacity, differing only in the activation of the physical prior, $\lambda = 0$ vs. $\lambda > 0$), *module-ablation* (where the ablation excises an entire multi-parameter neural block), or *both*; and (ii) *harm reporting*, classified as *reported* (explicitly documenting operating regimes or failure boundaries where physical constraints degrade downstream performance) or *not-reported*.

4.2. Multi-Site Operational Wind Farm Testbed and SCADA Telemetry

To ground the empirical benchmark in real-world power system operations across diverse terrains, we established an operational testbed across three commercial UK wind farms with 10-minute SCADA telemetry:

- **Kelmarsh Wind Farm** (Northamptonshire, England): Located in flat open agricultural terrain, comprising six Senvion MM92 turbines (rated 2.05 MW each, hub height 78 m, farm capacity 12.3 MW), evaluated over $N_{\text{train}} = 5,098$ clean intervals as our primary development benchmark (8,212 test origins, 49,272 forward delivery steps across full-year 2017) [25].

- **Hill of Towie Wind Farm** (Drummuir, Scotland): Located in complex upland terrain featuring micro-topographical wake interactions, comprising 21 Siemens SWT-2.3-82 turbines (rated 2.3 MW each, hub height 60 m, farm capacity 48.3 MW), evaluated as an independent spatiotemporal extrapolation benchmark across full-year 2017 with 7,599 forecast origins (45,594 delivery steps) [26].

- **Penmanshiel Wind Farm** (Scottish Borders): Locat-

ed in undulating upland terrain, comprising 14 Senvion MM82 turbines (rated 2.05 MW each, farm capacity 28.7 MW), configured with a sample-restricted training partition ($N_{\text{train}} = 1,583$) across 8,440 test origins to evaluate model behavior in this sample-restricted setting [27].

Standard data sanitation protocols removed periods of turbine curtailment, anemometer icing, and communication dropouts ($P \leq 0$ when $v > v_{\text{cut-in}}$ or active power frozen for > 6 consecutive steps). Real-world balancing price dynamics were evaluated using official 2017 UK Elexon BSC System Buy Prices (mean £44.27/MWh, peak £1,509.80/MWh, with 68 negative price periods).

4.3. The Controlled Attribution Protocol

To overcome the confounding inherent in standard benchmarking, we establish the **Controlled Attribution Protocol**. Rather than comparing a physics-constrained neural network against an unrelated data-driven baseline or excising neural submodules, this protocol holds all non-physical factors strictly invariant:

1. **Backbone and Parameter Parity**: All candidate models within each benchmark arm share identical neural architectures (DLinear, [28]), identical layer counts, and identical parameter weights.

2. **Data Splits and Optimization Schedules**: Training, validation, and test splits are strictly synchronized across all arms. Within each controlled comparison block, treatment and control arms used identical paired random seeds. Seed sets differed across distinct experiment families and are reported explicitly in the corresponding tables and supplementary experiment registry.

3. **Form- and Weight-Matched Pseudo-Scrambled Controls**: To separate authentic aerodynamic guidance from non-specific numerical smoothing, we construct negative controls that permute the empirical wind-to-power mapping along the wind speed axis while holding network architectures, loss formulations, and penalty weights λ 100% identical.

4. **Decoupled Post-Hoc Quantile Calibration**: All forecasters undergo identical post-hoc scalar quantile calibration ($\delta^* = \arg\min_\delta \hat{\mathcal{R}}_{\text{val}}(f_\theta + \delta)$ at $r = 2, q = 1/3$), evaluated strictly on the 2016 validation year before accessing the 2017 test timeline. This ensures that reported differences reflect genuine representation accuracy rather than uncalibrated quantile offsets.

4.4. Controlled Ablation Arms and Boundary Stress Tests

The forecasting suite evaluates an input lookback of $L = 36$ (6 hours) and a forecast horizon of $H = 36$ (6 hours), with downstream economic dispatch evaluated over the next-hour delivery window $H_{\text{disp}} = \{1, \ldots, 6\}$ and an origination stride of $S = 6$ steps to prevent overlapping settlement intervals. The experimental arms comprise:

1. *Unconstrained DLinear*: Baseline linear mapping across trend and seasonal decomposition ($\lambda = 0$) [28].

2. *Authentic Empirical Power Curve Prior*: Penalizing squared deviations from site-specific empirical power curves across penalty weights $\lambda \in \{0.01, 0.1, 1.0\}$; during test inference, the model operates purely on historical power telemetry ($\mathcal{D}_3 = \texttt{none}$).

3. *Form- and Weight-Matched Pseudo-Scrambled Prior*: Permuting the aerodynamic power curve mapping across identical $\lambda \in \{0.01, 0.1, 1.0\}$.

4. *Info-Symmetric Reference Arm*: Supplying contemporaneous wind speed directly as an unconstrained input feature to assess whether symmetric error reductions align with operational dispatch risk.

5. *Strictly Budget-Matched Information Ablation* ($N_{\text{prior}} = N_{\text{train}} = 500$): To decouple domain prior geometry from historical sample expansion, models trained on restricted windows ($N_{\text{train}} = 500$, ≈ 21 days) are evaluated against both budget-matched priors (fitted strictly on the identical 500 samples) and full-history priors (fitted on 7–10 months).

6. *Within-Site Learning Curves*: Subsampling training volume across $N_{\text{train}} \in \{500, 1000, 1500, 3000, 5000\}$ on Kelmarsh and Hill of Towie across five random seeds.

7. *Continuous Prior Corruption and Engineering Drift*: Continuously interpolating between authentic and scrambled priors ($\alpha \in [0.0, 1.0]$), simulating aerodynamic aging via rated capacity derating (0% to 20%), and applying anemometer calibration drift (−15% to +15%).

8. *Training Proxy Staleness Scan*: Scanning historical wind speed proxy lag $k \in \{6, 12, 18, 36\}$ steps (1h to 6h delay) in the training loss objective.

9. *Cross-Backbone Architectural Probe (PatchTST)*: Evaluating PatchTST ([29], patch length 6, stride 4, 2-layer encoder with RevIN) across 60 independently trained models (3 farms $\times$ 4 constraints $\times$ 5 seeds) under nominal regularization ($\lambda = 0.01$).

Statistical significance was evaluated via 2,000-replication circular block paired bootstrapping with 7-day block lengths ($L = 1,008$ intervals) on full-year continuous telemetry (Kelmarsh and Hill of Towie) to account for meteorological autocorrelation, combined with multi-seed standard deviations to capture optimization variance across the unified five seeds.

4.5. Governing Attribution Scope: What Is Evaluated vs. What Is Not

To ensure that empirical observations are not conflated with broader causal or mechanistic claims, we explicitly define the formal attributional scope of our experimental design:

Empirically Distinguished Under the Controlled Protocol.

(a) *Prior-Specific Inductive Guidance vs. Generic Regularization*: Under the tested matched-control protocol, divergence between authentic and pseudo-scrambled priors provides evidence that the observed performance shift is specific to the empirical aerodynamic mapping, rather than being explained by penalty form and weight alone.

(b) *Independence from Historical Volume Expansion*: The 500-sample budget-matched control isolates whether empirical benefits persist when prior fitting is restricted strictly to the contemporaneous training window, bounding the influence of historical telemetry volume.

(c) *Empirical Operating Boundaries*: Parametric scans map empirical sensitivity to sample size attenuation ($N_{\text{train}} \in \{500, \ldots, 5000\}$), continuous prior corruption crossover regions (α), aerodynamic derating, and training-loss proxy staleness (k) within the tested configurations. These values are empirical crossover locations along the evaluated perturbation grid, not population-level physical thresholds.

Not Formally Identified (Confounding Disclaimers).

(i) *Abstract Causal Effects of Physics*: Our experiments evaluate specific empirical power-curve and distributional constraints under evaluated configurations; they do not establish universal causal claims for "physics" as an abstract treatment across arbitrary architectures or energy systems.

(ii) *Cross-Site Topographical Causation*: Differences across Kelmarsh, Hill of Towie, and Penmanshiel confound topographic complexity with turbine ratings (2.05 MW vs. 2.30 MW), hub heights, wake geometry, and local micro-climates; topography remains an observational covariate rather than an isolated treatment.

(iii) *Direct Bias–Variance Component Estimation*: Our theoretical framework provides a decision-theoretic representation/calibration/cross-coupling decomposition; we do not directly estimate classical statistical bias and variance scalars from empirical data.

5. Empirical Centerpiece: When Does a Physical Prior Provide Genuine Value?

This section presents the core empirical findings of our Controlled Attribution Protocol. Across three operational commercial wind farms, we evaluate prior-specific mapping effects in this sample-restricted setting, evaluate weak-regularization numerical mimicry, and decouple domain geometric structure from auxiliary historical data volume under matched model capacities.

5.1. Prior Specificity under Sample Restriction

On the Penmanshiel wind farm evaluated under a sample-restricted training regime ($N_{\text{train}} = 1{,}583$), the unconstrained DLinear model exhibits empirical overfitting, yielding a normalized asymmetric dispatch loss of 0.09852 ± 0.00018 ($\text{Cost}_{\text{static}}$ at $r = 2, q = 1/3$; Table 2, Figure 2a). Embedding the authentic empirical power curve prior under strong regularization ($\lambda = 1.0$) reduces normalized dispatch cost to 0.09726 ± 0.00022 (a -1.28% risk reduction).

In sharp contrast, evaluating the form- and weight-matched Pseudo-Scrambled negative control—sharing the identical DLinear architecture, data split, training seed, and penalty multiplier ($\lambda = 1.0$), but permuting the power output values along the wind speed axis—inflates dispatch cost to 0.15884 ± 0.01588 (a $+61.2\%$ surge). Because the scrambled control preserves identical gradient capacity and loss formulation, this divergence demonstrates that under strong regularization in sample-restricted regimes, the model specifically benefits from authentic aerodynamic domain structure: valid domain structure provides measurable downside protection, whereas an unaligned mapping inflicts severe operational penalties.

Table 2: Controlled Ablation Benchmark with Explicit Penalty Conditions: Prior Specificity, Weak-Regularization Mimicry, and Training-Proxy Lag Across Three Operational Wind Farms (Quantile Calibrated $r = 2, q = 1/3$).

Wind Farm & Topology	Controlled Experimental Arm (Ablation Condition)	Normalized Cost $\text{Cost}_{\text{static}}$ ($r = 2$)	Pinball Loss ($q = 1/3$)	RMSE (pu)	Shortfall (£/(turb·yr))	Empirical Takeaway & Diagnostic Assessment
Kelmarsh (12.3 MW, 6 turb) *Flat Open Terrain* $N = 5{,}098$	Vanilla Unconstrained ($\lambda = 0$)	0.08804 ± 0.00006	0.02935	0.1717	£5,764.03	Pure-data power autoregressive baseline
	Info-Symmetric (Power + Wind)	0.08923 ± 0.00051	0.02975	0.1696	£5,979.36	Wind speed reduces RMSE but elevates tail dispatch cost
	Genuine Curve Prior ($\lambda = 0.1$)	0.08785 ± 0.00005	**0.02928**	0.1707	£5,726.07	Modest observed risk reduction (−0.22%)
	Pseudo-Scrambled Prior ($\lambda = 0.01$)	0.08792 ± 0.00008	0.02931	0.1729	£5,492.15	**Regularization Mimicry**: matches genuine prior (0.08803 at $\lambda = 0.01$)
	Genuine Distribution ($\lambda = 0.1$)	0.09810 ± 0.00077	0.03270	0.1701	£8,106.35	Shortfall penalty surges +40.6% under distribution constraint
	Stale Training Proxy ($k = 36$, 6h)	0.09034 ± 0.00005	0.03011	0.1729	£5,807.85	Monotonic degradation as lagged wind speed proxy decays
Hill of Towie (48.3 MW, 21 turb) *Complex Upland* $N = 7{,}035$	Vanilla Unconstrained ($\lambda = 0$)	0.08913 ± 0.00005	0.02971	0.1740	£5,941.22	Robust unconstrained benchmark
	Info-Symmetric (Power + Wind)	0.09028 ± 0.00017	0.03009	0.1733	£6,422.95	Wind channel adds high-frequency variance to dispatch
	Genuine Curve Prior ($\lambda = 0.1$)	0.08905 ± 0.00004	**0.02968**	0.1745	£5,917.69	Modest observed risk reduction (−0.09%) under moderate penalty
	Pseudo-Scrambled Prior ($\lambda = 0.01$)	0.08910 ± 0.00005	0.02970	0.1740	£6,003.64	Closely matched to genuine prior under tested weak-penalty setting (0.08913 at $\lambda = 0.01$)
	Genuine Distribution ($\lambda = 0.1$)	0.09445 ± 0.00041	0.03148	0.1800	£5,824.12	**Metric Inversion**: cashflow rises while decision cost degrades
	Stale Training Proxy ($k = 36$, 6h)	0.09246 ± 0.00006	0.03082	0.1759	£6,174.00	Temporal proxy lag monotonically erodes prior guidance
Penmanshiel (28.7 MW, 14 turb) *Complex Upland* $N = 1{,}583$ *(Sample-Restricted)*	Vanilla Unconstrained ($\lambda = 0$)	0.09852 ± 0.00018	0.03284	0.2059	£6,658.86	Sample-restricted training partition prone to empirical over-fitting
	Info-Symmetric (Power + Wind)	0.10360 ± 0.00078	0.03453	0.2090	£6,717.80	Additional unconstrained input channels exacerbate over-fitting
	Genuine Curve Prior ($\lambda = 1.0$)	0.09726 ± 0.00022	**0.03242**	0.2054	£6,682.66	**Authentic Inductive Benefit** (−1.28%) in sample-restricted regime
	Pseudo-Scrambled Prior ($\lambda = 1.0$)	0.15884 ± 0.01588	0.05295	0.2386	£11,692.72	**Prior Specificity Demonstration**: scrambling mapping triggers +61.2% decision loss surge
	Genuine Distribution ($\lambda = 0.01$)	0.09795 ± 0.00022	0.03265	0.2056	£6,500.65	Modest regularization under calibrated penalty
	Stale Training Proxy ($k = 36$, 6h)	0.10425 ± 0.00025	0.03475	0.2083	£6,981.00	Lag decay completely reverses prior advantage by Step 36

Notes: (1) Values report multi-seed mean ± seed standard deviation across clean operational test periods across 5 random seeds (42, 123, 456, 789, 1024). Penalty schedules ($\lambda \in \{0.01, 0.1, 1.0\}$) were defined across a predefined sensitivity and stress-testing grid; model checkpoints and calibration offsets were selected exclusively on the 2016 validation/calibration partition, with the 2017 test year accessed only after protocol freezing. (2) $\text{Cost}_{\text{static}} = 3 \times \text{PinballLoss}(q = 1/3)$ under post-hoc quantile calibration ($r = 2$). (3) Pseudo-Scrambled arms permute the empirical wind-to-power mapping while preserving identical loss formulations and hyperparameters λ. At weak penalty ($\lambda = 0.01$), genuine curve and scrambled curve yield closely matched outcomes with no clear empirical separation (0.08803 vs. 0.08792 on Kelmarsh; 0.08913 vs. 0.08910 on Hill of Towie), consistent with non-specific numerical regularization. At strong penalty ($\lambda = 1.0$) on Penmanshiel, genuine curve reduces loss (−1.28%) whereas scrambled prior inflates loss (+61.2%), isolating prior-specific mapping effects in this sample-restricted setting. (4) Stale Training Proxy evaluates sensitivity to historical wind speed proxy lag $k \in \{6, 12, 18, 36\}$ (1h to 6h) in the training objective; the model takes only historical power at inference.

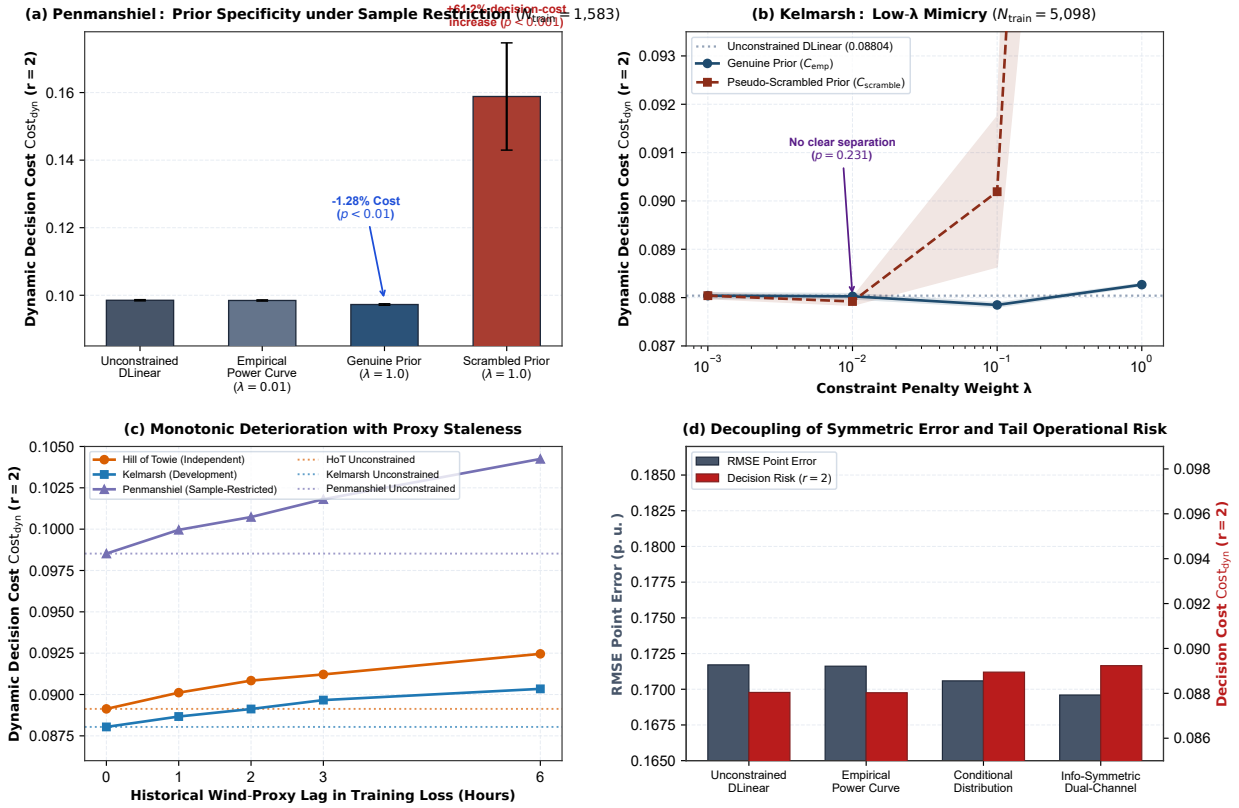


Figure 2: Prospective Controlled Attribution Benchmark across Three Operational Wind Farms. (a) Prior-specific mapping effects under strong regularization on Penmanshiel (N_{train} = 1,583): authentic empirical prior guidance provides downside risk reduction (−1.28%), whereas permuting the aerodynamic mapping (Pseudo-Scrambled prior) triggers a +61.2% surge in operational dispatch cost. (b) Weak-penalty numerical regularization mimicry on Kelmarsh (N = 5,098), where genuine and scrambled priors yield closely matched outcomes at λ = 0.01, with no clear empirical separation and outcomes consistent with non-specific numerical regularization. (c) Monotonic cost inflation under training proxy staleness: dispatch cost escalation across all three farms as historical wind proxy lag k in the training loss increases from 1 h to 6 h. (d) Decoupling of symmetric point error (RMSE) from asymmetric decision risk (r = 2), demonstrating that point metrics mask tail operational hazards.

5.2. *Weak-Penalty Scale: Non-Specific Numerical Regularization Mimicry*

A prevalent assertion in the physics-informed literature is that minor performance gains observed under weak penalty weights (e.g., $\lambda \leq 0.01$) demonstrate that deep networks internalize physical domain principles. Our Controlled Attribution Protocol directly interrogates this premise.

At weak regularization (λ = 0.01) on Kelmarsh (N_{train} = 5,098), the genuine physical power curve prior yields a normalized dispatch cost of 0.08803 ± 0.00005, while the form-matched pseudo-scrambled control yields 0.08792±0.00008 (Table 2, Figure 2b). On Hill of Towie (N_{train} = 7,035), genuine and scrambled priors yield 0.08913 ± 0.00005 and 0.08910 ± 0.00005, respectively. Across both wind farms, genuine and scrambled priors yield closely matched outcomes with no clear empirical separation (differences < 0.0001, within multi-seed error margins).

Because structurally permuted negative controls reproduce the empirical outcomes of genuine aerodynamics under weak penalty scaling, the observed effect cannot be specifically attributed to the physical mapping and remains consistent with non-specific numerical regularization. This result provides a critical negative control: minor accuracy gains under weak penalty multipliers should not be cited as evidence of domain physical reasoning without form-matched permutation controls.

5.3. *Information-Budget Alignment: Decoupling Domain Geometry from Data Volume*

A central threat to the attribution of physical priors in low-sample regimes is the *information budget discrepancy*: when an empirical prior is fitted on an extensive historical archive (e.g., 7 to 10 months of SCADA

telemetry), does its advantage stem from domain aerodynamic geometry or from the unacknowledged volume of historical operational data?

To resolve this question, we implemented a strictly budget-matched information control at $N_{\text{train}} = 500$ (≈ 21 consecutive days of telemetry; Table 3, all prior arms $\lambda = 1.0$, five random seeds). Models trained on this restricted window were evaluated against: (i) an unconstrained baseline ($\lambda = 0$); (ii) a budget-matched genuine prior fitted strictly on the exact same 500 training samples ($N_{\text{prior}} = N_{\text{train}} = 500$); (iii) a full-historical genuine prior fitted on the multi-month record; (iv) a budget-matched scrambled control; and (v) a full-historical scrambled control.

As detailed in Table 3, the unconstrained baseline yields a normalized dispatch cost of 0.09648 ± 0.00314 on Kelmarsh and 0.10444 ± 0.00312 on Hill of Towie. Introducing the strictly budget-matched genuine prior ($N_{\text{prior}} = 500$) reduces dispatch cost to 0.09499 ± 0.00307 (-1.54%) on Kelmarsh and 0.10095 ± 0.00305 (-3.34%) on Hill of Towie. The budget-matched genuine prior retains approximately 77% of the full-history mean benefit on Kelmarsh (-1.54% vs. -2.00%) and approximately 92% on Hill of Towie (-3.34% vs. -3.62%).

Evaluated under paired differences across the five identical random seeds, the budget-matched genuine prior achieves a statistically significant mean paired risk reduction of -0.00149 (95% CI: $[-0.00174, -0.00124]$, $p < 0.01$) on Kelmarsh and -0.00349 (95% CI: $[-0.00368, -0.00331]$, $p < 0.01$) on Hill of Towie relative to unconstrained baselines. Expanding the prior fitting set to the full historical record yields an incremental reduction of only -0.00044 (95% CI: $[-0.00075, -0.00013]$) on Kelmarsh and -0.00029 (95% CI: $[-0.00041, -0.00016]$) on Hill of Towie. In contrast, evaluating the budget-matched scrambled control inflates dispatch cost to 0.13827 ± 0.02333 (+43.3%) on Kelmarsh and 0.15248 ± 0.00253 (+46.0%) on Hill of Towie.

These findings provide strong controlled evidence that **the majority of the observed benefit cannot be explained by additional historical data volume alone**. The algebraic and geometric structure of the empirical power curve provides genuine inductive guidance in this sample-restricted setting, even when estimated strictly from the limited contemporaneous training sample.

6. Mapping the Operating Envelope within the Evaluated Settings: Where Does the Advantage Break?

This section maps the empirical operating envelope of physical priors within the evaluated settings, evaluating boundary conditions across training sample volume, prior corruption, engineering drift, temporal proxy misalignment, and neural backbone architectures.

6.1. Within-Site Learning Curves across Sample Budgets

To decouple sample volume from geographic topography and examine whether physical prior advantages persist under data expansion, we evaluated within-site learning curves across a discrete grid of training sample budgets $N_{\text{train}} \in \{500, 1000, 1500, 3000, 5000\}$ on Kelmarsh and Hill of Towie across five random seeds (Figure 3).

In the tested sample-restricted regimes ($N_{\text{train}} = 500$ and $N_{\text{train}} = 1{,}000$), embedding the authentic empirical power curve prior ($\lambda = 1.0$) provides clear downside risk protection:

- At $N_{\text{train}} = 500$, normalized dispatch cost decreases from 0.09648 ± 0.00314 to 0.09455 ± 0.00285 (-2.00%) on Kelmarsh, and from 0.10444 ± 0.00312 to 0.10066 ± 0.00307 (-3.62%) on Hill of Towie.
- At $N_{\text{train}} = 1{,}000$, the genuine prior maintains an advantage on both sites (0.09241 vs. 0.09365, -1.32% on Kelmarsh; 0.09681 vs. 0.09902, -2.23% on Hill of Towie).

However, as training volume increases across the intermediate budgets ($N_{\text{train}} = 1{,}500$ and $3{,}000$), the margin of advantage steadily attenuates. At the largest evaluated training budget ($N_{\text{train}} = 5{,}000$), unconstrained DLinear achieves a dispatch cost of 0.08812 ± 0.00022 on Kelmarsh and 0.08936 ± 0.00023 on Hill of Towie, closing the gap and slightly outperforming the fixed genuine prior (0.08841 ± 0.00039, +0.33% on Kelmarsh; 0.09091 ± 0.00131, +1.73% on Hill of Towie). Throughout all evaluated sample sizes, form-matched scrambled negative controls inflict severe performance degradations (+43% to +98%), confirming that unaligned priors remain destructive regardless of sample availability.

Conceptual Interpretation. The observed learning curves are consistent with a conceptual hypothesis in which the regularizing value of variance reduction diminishes as sample availability increases, while structural approximation bias becomes increasingly consequential. Under this interpretive perspective, in sample-restricted windows, unconstrained models are hypothesized to be vulnerable to elevated estimation variance that an empirical power curve helps suppress by constraining the hypothesis space. Conversely, purely data-driven representations can fit the observed input–output relationship

Table 3: Controlled Robustness Ablation under Matched Sample Windows at $N_{\text{train}} = 500$ ($\approx$ 21 Days Training Telemetry): Disentangling Domain Prior Geometry from Auxiliary Historical Data Budgets (Quantile Calibrated $r = 2, q = 1/3$, All Prior Arms $\lambda = 1.0$, 5 Seeds).

Wind Farm & Topology	**Controlled Experimental Arm Condition**	**Model Training Budget**	**Prior Information Budget**	**Cost ($r = 2$) (Mean ± Std)**	**Empirical Takeaway & Identification Rationale**
Kelmarsh (12.3 MW) *Flat Open Terrain*	A. No Physical Prior ($\lambda = 0$)	500 samples	None	0.09648 ± 0.00314	Pure-data unconstrained baseline under data restriction
	B. Budget-Matched Genuine Prior ($\lambda = 1.0$)	500 samples	500 samples	0.09499 ± 0.00307	**Domain Structure Effect** (−1.54% mean): retains ≈ 77% of full-history mean gain
	C. Full-Historical Genuine Prior ($\lambda = 1.0$)	500 samples	7 months (full)	0.09455 ± 0.00285	Full-history mean gain (−2.00%): historical pool adds modest incremental gain (−0.46%)
	D. Budget-Matched Scrambled Control ($\lambda = 1.0$)	500 samples	500 samples	0.13827 ± 0.02333	**Prior Specificity**: permuting mapping triggers +43.3% cost inflation
	E. Full-Historical Scrambled Control ($\lambda = 1.0$)	500 samples	7 months (full)	0.13798 ± 0.02338	Scrambling across full seasonal history remains severely harmful (+43.0%)
Hill of Towie (48.3 MW) *Complex Upland*	A. No Physical Prior ($\lambda = 0$)	500 samples	None	0.10444 ± 0.00312	Sample-restricted unconstrained baseline prone to empirical overfitting
	B. Budget-Matched Genuine Prior ($\lambda = 1.0$)	500 samples	500 samples	0.10095 ± 0.00305	**Domain Structure Effect** (−3.34% mean): retains ≈ 92% of full-history mean gain
	C. Full-Historical Genuine Prior ($\lambda = 1.0$)	500 samples	10 months (full)	0.10066 ± 0.00307	Full-history mean gain (−3.62%): historical pool adds modest incremental gain (−0.28%)
	D. Budget-Matched Scrambled Control ($\lambda = 1.0$)	500 samples	500 samples	0.15248 ± 0.00253	**Prior Specificity**: permuting mapping triggers +46.0% cost inflation
	E. Full-Historical Scrambled Control ($\lambda = 1.0$)	500 samples	10 months (full)	0.17209 ± 0.01703	Severe misspecification penalty under strong constraint (+64.8%)

Notes: (1) All models share identical DLinear architectures, optimization parameters, and post-hoc quantile calibration offsets ($r = 2, q = 1/3$) evaluated over the full-year 2017 test timeline across 5 random seeds (42, 123, 456, 789, 1024). All genuine and scrambled prior arms (Arms B–E) evaluate under penalty weight $\lambda = 1.0$. (2) Arms A, B, and D share the exact same 500-sample training data window ($\approx$ 21 consecutive days). The budget-matched prior retains most of the mean improvement observed with the full-history prior (approximately 77% on Kelmarsh and 92% on Hill of Towie). Under paired differences across the 5 identical seeds, the budget-matched prior yields a mean paired risk reduction of −0.00149 (95% CI: [−0.00174, −0.00124]) on Kelmarsh and −0.00349 (95% CI: [−0.00368, −0.00331]) on Hill of Towie relative to Arm A, while expanding the prior to full historical records provides an incremental paired reduction of −0.00044 (95% CI: [−0.00075, −0.00013]) and −0.00029 (95% CI: [−0.00041, −0.00016]), respectively. (3) These paired comparisons indicate that domain-structured empirical prior effects under a matched temporal data window account for the majority of the observed risk reduction, rather than extra historical data volume alone.

more flexibly as training coverage expands; under these well-sampled conditions, a rigid, static prior can act as a structural approximation bias that constrains the model from fitting nuanced aerodynamic interactions. We emphasize that this bias–variance trade-off is a conceptual interpretation consistent with—not formally identified or proven by—the empirical learning curves, as scalar bias and variance components were not directly decomposed or measured from empirical data.

Joint Operating Envelope. Taken together, the learning-curve experiments (at $\lambda = 1.0$) and separate penalty-sensitivity evaluations suggest that both data availability and regularization strength govern the operating envelope. In the absence of a dense factorial grid across multiple $N \times$ multiple λ values, we characterize this dependency as a joint empirical tendency rather than a statistically proven interaction.

6.2. Continuous Prior Corruption and Engineering Drift

To test whether prior vulnerability is restricted to total permutation, we evaluated continuous prior perturbation and engineering drift models on Penmanshiel (Figure 4).

Continuous Corruption Interpolation ($\alpha \in [0.0, 1.0]$). We constructed a continuous perturbation trajectory pa-

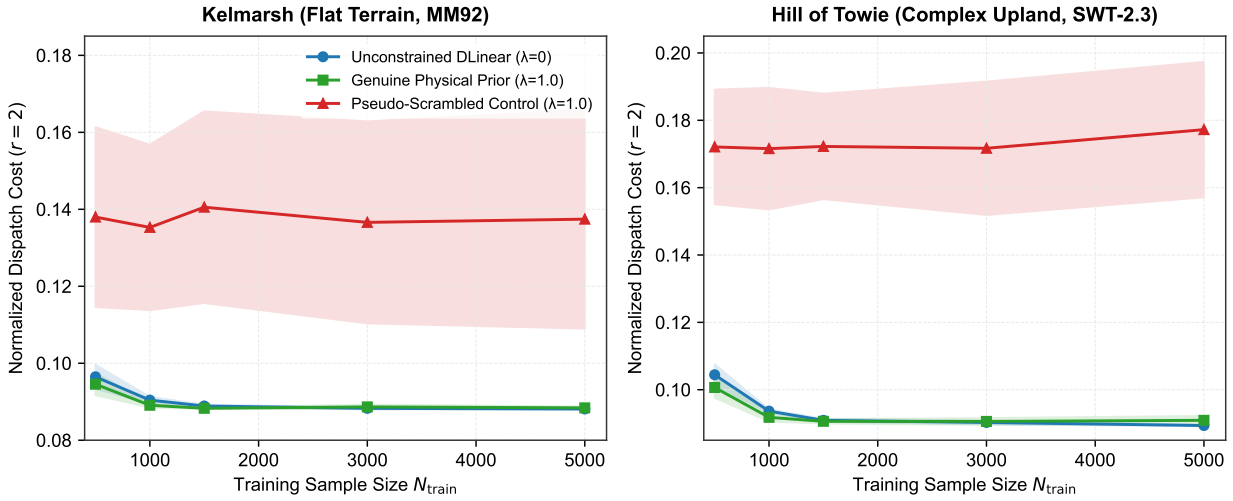


Figure 3: Within-Site Learning Curves across Discrete Sample Sizes $N_{\text{train}} \in \{500, 1000, 1500, 3000, 5000\}$ on Kelmarsh and Hill of Towie (Multi-seed means and standard deviations). Holding site topography and turbine hardware constant, authentic empirical priors yield lower dispatch costs in the tested sample-restricted regimes ($N_{\text{train}} \in \{500, 1{,}000\}$, -2.00% on Kelmarsh and -3.62% on Hill of Towie at $N = 500$). As training volume expands, this advantage attenuates; at the largest evaluated training budget ($N_{\text{train}} = 5{,}000$), unconstrained data-driven models close the gap and slightly reverse the ranking ($+0.33\%$ on Kelmarsh, $+1.73\%$ on Hill of Towie). In contrast, form-matched scrambled priors under strong regularization ($\lambda = 1.0$) induce severe dispatch cost penalties ($+43\%$ to $+98\%$) across all evaluated sample regimes.

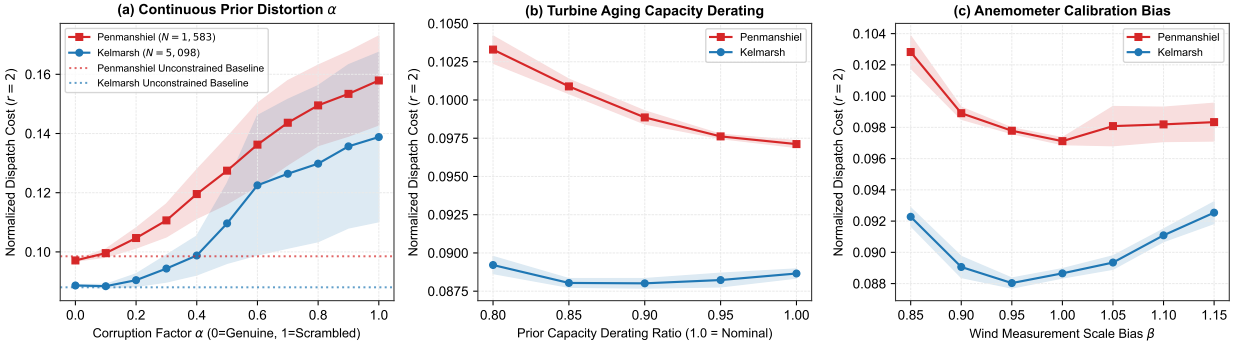


Figure 4: Continuous Prior Corruption and Engineering Drift Severity Response Curves. (a) Continuous interpolation between authentic and scrambled priors $\alpha \in [0.0, 1.0]$: on Penmanshiel, the tested corruption trajectory crosses the unconstrained baseline around an empirical crossover region of $\alpha \approx 0.07$. (b) Turbine aerodynamic aging: capacity derating exceeding 10% flips nominal physical priors into economic penalties under the evaluated configuration. (c) Anemometer sensor calibration drift: Penmanshiel retains an advantage under positive wind speed drift up to +15%, whereas negative sensor under-reporting systematically inflates shortfall.

rameterized by $\alpha \in [0.0, 1.0]$, linearly blending the authentic power curve with its scrambled permutation. As shown in Figure 4a, dispatch cost degrades monotonically with perturbation severity. On Penmanshiel, the tested corruption trajectory crosses the unconstrained baseline around an empirical crossover region of $\alpha \approx 0.07$: costs rise from 0.09712 ± 0.00031 at $\alpha = 0.0$ to 0.09958 ± 0.00042 at $\alpha = 0.1$. At $\alpha = 0.2$, normalized dispatch cost reaches 0.10469 ± 0.00338 (a +5.9% degradation relative to unconstrained models). We emphasize that these values are empirical crossover locations along the evaluated perturbation grid, not population-level physical thresholds, indicating that even modest distortion (under 10% permutation blend) is sufficient to erode inductive benefits in this configuration.

Turbine Aerodynamic Aging and Derating. In operational wind fleets, blade erosion, leading-edge roughness, and mechanical aging cause progressive aerodynamic derating. In the evaluated Penmanshiel configuration, simulating turbine capacity derating reveals that derating exceeding 10% flips nominal physical priors into net eco-

nomic penalties (Figure 4b), as the network regularizes against an un-derated historical envelope that exceeds the turbine's degraded capability. This 10% threshold reflects the evaluated operating margin where representation bias overtakes variance reduction.

Anemometer Calibration Drift. Simulating anemometer measurement drift (±15%) reveals asymmetric operational tolerances (Figure 4c). On Penmanshiel, the model retains a modest advantage over unconstrained baselines under positive wind speed drift up to +15% (0.09833 vs. 0.09852). In contrast, negative wind speed underreporting systematically depresses predicted power, triggering substantial shortfall penalties during high-wind ramp events.

6.3. Sensitivity to Training-Loss Proxy Staleness

Many physical constraint formulations rely on meteorological proxies (such as supervisory anemometer readings) to query empirical power curves during loss evaluation. To assess sensitivity to proxy latency in the absence of forward numerical weather predictions, we performed a parametric lag scan across $k \in \{6, 12, 18, 36\}$ steps (representing 1 h to 6 h delay in the training proxy; Figure 2c).

In the Hill of Towie lag scan, decision risk exceeds the unconstrained reference (0.08913) by the first evaluated lag $k = 6$ (1 h delay, normalized dispatch cost 0.09011 ± 0.00008) and deteriorates monotonically through $k = 36$ (0.09246 ± 0.00006). On Kelmarsh, dispatch cost rises from 0.08867 ($k = 6$) to 0.09034 ($k = 36$). On Penmanshiel, dispatch cost escalates from 0.09996 ($k = 6$) to 0.10425 ($k = 36$). Because $k = 6$ is the minimum non-zero lag evaluated in this discrete grid, this observation establishes that latency begins degrading operational value within the first hour of proxy delay, while more granular sub-hour transitions remain unobserved.

This experiment isolates the sensitivity of offline training objectives to temporal misalignment between the forecast horizon and the historical weather proxy used to index the power curve. It demonstrates that when contemporaneous or forward meteorological forecasts are unavailable, regularizing predictions against stale weather proxies monotonically degrades downstream dispatch performance.

6.4. Cross-Backbone Architectural Sensitivity Probe (PatchTST)

To examine whether boundary behaviors are specific to linear decomposition architectures, we evaluated PatchTST [29] across 60 independently trained models (3 farms × 4 constraint conditions × 5 random seeds; Table 4) under nominal regularization ($\lambda = 0.01$).

As detailed in Table 4, PatchTST confirms that distribution priors consistently reduce dynamic shortfall expenditure across all three wind farms (saving £232.44, £127.49, and £790.51/(turb·yr) on Hill of Towie, Kelmarsh, and Penmanshiel, respectively), indicating that defensive shortfall reduction is preserved in non-linear patch representations. However, net balancing cashflow exhibits a sharp architectural divergence: on Hill of Towie, distribution regularization decreases net cashflow by £568.68/(turb·yr) under PatchTST, inverting the positive cashflow increment observed under DLinear.

Furthermore, on Penmanshiel in the sample-restricted regime, simple empirical baselines (Ridge regression at 0.09744 and Persistence at 0.09586) outperform all constrained deep networks (PatchTST at 0.11134). This confirms that external physical loss penalties cannot uniformly compensate for deep parameter overfitting when training data is severely restricted.

7. Decision Alignment and Supporting Software Forensics

This section investigates why conventional evaluation metrics and software assumptions often mislead. We demonstrate that symmetric point accuracy and standalone balancing cashflows decouple from operational decision risk, and present supporting forensic evidence from computational graph auditing.

7.1. Decision-Metric Decoupling: The Info-Symmetric Benchmark

A widespread assumption in energy forecasting is that minimizing symmetric point regression error (RMSE or MAE) reliably translates into superior operational decisions. Our Controlled Attribution Protocol provides a clean empirical refutation of this assumption through the **Info-Symmetric arm** (Table 2).

On the Kelmarsh wind farm, the unconstrained baseline operating solely on historical power inputs achieves an RMSE of 0.1717 and a normalized asymmetric dispatch cost of 0.08804 ± 0.00006 ($r = 2, q = 1/3$). Introducing contemporaneous wind speed telemetry as an additional input feature (the Info-Symmetric arm) improves point prediction accuracy, reducing RMSE from 0.1717 to 0.1696 (a 1.22% improvement). However, this symmetric error reduction concurrently **worsens** the downstream operational dispatch cost from 0.08804 to

Table 4: Exploratory Cross-Backbone Architectural Sensitivity Analysis: Deep Transformer (PatchTST) under Half-Hour Imbalance Cashflow Accounting (5-Seed Means; Multi-Seed Dispersions Documented in Table Notes).

Site	PatchTST Model Variant	Policy	RMSE (pu)	Normalized Dynamic Cost Cost_{dyn}	Shortfall Cashout (£/(turb·yr))	Balancing Imbalance Cashflow (£/(turb·yr))
Hill of Towie (48.3 MW, Upland)	Unconstrained PatchTST	Uncalibrated	0.1176	0.10518	£21,753.77	£12,292.64
	Unconstrained PatchTST	**Calibrated**	0.1202	0.10220	£18,168.11	£20,822.29
	Power Curve ($\lambda = 0.01$)	Calibrated	0.1201	0.10212	£18,163.56	£20,765.01
	Distribution Prior ($\lambda = 0.01$)	Uncalibrated	0.1162	0.10507	£22,362.12	£9,762.57
	Distribution Prior ($\lambda = 0.01$)	**Calibrated**	0.1192	0.10129	**£17,935.67**	**£20,253.61**
Kelmarsh (12.3 MW, Flat)	Unconstrained PatchTST	Uncalibrated	0.1092	0.10889	£22,630.77	£4,523.36
	Unconstrained PatchTST	**Calibrated**	0.1118	0.10222	**£16,808.90**	**£17,411.89**
	Power Curve ($\lambda = 0.01$)	Calibrated	0.1117	0.10212	£16,774.31	£17,431.54
	Distribution Prior ($\lambda = 0.01$)	Uncalibrated	0.1075	0.10870	£23,336.46	£1,528.06
	Distribution Prior ($\lambda = 0.01$)	**Calibrated**	0.1097	0.10075	£16,681.41	£16,046.77
Penmanshiel (28.7 MW, Upland)	Unconstrained PatchTST	Uncalibrated	0.1268	0.12107	£26,956.48	£3,001.45
	Unconstrained PatchTST	**Calibrated**	0.1304	0.11270	£19,129.15	£20,453.24
	Power Curve ($\lambda = 0.01$)	Calibrated	0.1303	0.11264	£19,118.38	£20,444.41
	Distribution Prior ($\lambda = 0.01$)	Uncalibrated	0.1265	0.11729	£24,606.87	£6,473.07
	Distribution Prior ($\lambda = 0.01$)	**Calibrated**	0.1300	0.11134	**£18,338.64**	**£21,117.53**

Notes: (1) Exploratory sensitivity results for PatchTST (Nie et al., ICLR 2023; patch length 6, stride 4, channel-independent RevIN) evaluate capacity-aligned architectural sensitivity under identical half-hour imbalance cashflow accounting across five random initialization seeds under nominal regularization $\lambda = 0.01$. At flat Kelmarsh, unconstrained PatchTST secures higher balancing cashflow (+£1,365.12/(turb·yr) over distribution prior); at upland Penmanshiel and Hill of Towie, distribution regularization provides modest downside risk reduction, saving £790.51/(turb·yr) and £232.44/(turb·yr) in single-sided shortfall. (2) Multi-seed standard deviations across 5 seeds: on Hill of Towie, calibrated unconstrained RMSE is 0.1202 ± 0.0009, Cost_{dyn} is 0.10220 ± 0.00072, and shortfall is £$18,168.11 \pm 438.72$; on Kelmarsh, calibrated unconstrained RMSE is 0.1118 ± 0.0019, Cost_{dyn} is 0.10222 ± 0.00172, and shortfall is £$16,808.90 \pm 777.10$; on Penmanshiel, calibrated unconstrained RMSE is 0.1304 ± 0.0013, Cost_{dyn} is 0.11270 ± 0.00133, and shortfall is £$19,129.15 \pm 856.32$.

0.08923 ± 0.00051 (a 1.35% cost inflation), while increasing annual shortfall expenditure from £5,764.03 to £5,979.36/(turb·yr).

Hill of Towie exhibits the identical qualitative direction: adding wind speed telemetry improves RMSE from 0.1740 to 0.1733, yet worsens operational dispatch cost from 0.08913 ± 0.00005 to 0.09028 ± 0.00017, driving shortfall expenditure from £5,941.22 to £6,422.95/(turb·yr).

This empirical divergence provides concrete real-world evidence that:

> **Improved point prediction does not necessarily imply improved operational decisions.**

Because symmetric metrics treat positive and negative errors identically, a model can achieve a lower global RMSE by reducing small errors across benign operating regimes, while simultaneously exacerbating tail under-prediction errors during high-cost shortfall events.

7.2. Balancing Cashflow Pathology: The Zero-Bid Reductio ad Absurdum

A second common evaluation practice is to score forecasting models on standalone unhedged balancing market cashflows ($R_{\text{imb}} = P_{\text{imb}} \cdot (y - q)$). Our experiments reveal that this practice contains a severe structural flaw.

Under half-hour settlement aggregation with 2017 UK BSC cash-out prices, evaluating a degenerate **Zero-Bid baseline** ($q \equiv 0$, submitting a forecast of zero generation across all settlement intervals) demonstrates this pathology (Table 5, Figure 5a).

Because $q \equiv 0 \leq y_t$ holds almost surely for non-negative wind generation, the Zero-Bid strategy incurs exactly **£0.00 in shortfall penalties** across all three wind farms. In standalone unhedged balancing accounting, Zero-Bid mechanically generates massive apparent cashflows:

- **Kelmarsh**: +£223,882/(turb·yr);
- **Hill of Towie**: +£239,299/(turb·yr);
- **Penmanshiel**: +£243,450/(turb·yr).

We explicitly emphasize that **these six-figure sums represent uncontracted physical spill, not commercial profit or economic value**. In actual wholesale electricity markets (governed by the two-settlement identity $R_{\text{total}} = P_{\text{F}} \cdot q + P_{\text{imb}} \cdot (y - q)$ in Eq. (13)), bidding zero forfeits all forward contract revenue ($P_{\text{F}} \cdot q = 0$). When evaluated under the ground-truth asymmetric operational dispatch loss ($r = 2, q = 1/3$), Zero-Bid is heavily penalized, yielding normalized decision costs of 0.30409 to 0.33689 (matching the normalized mean capacity factor $\bar{y}$, Table 5).

Table 5: Market Settlement Diagnostic Audit: Exposing the Pitfalls of Standalone Imbalance Cashflow via the Degenerate Zero-Bid ($q \equiv 0$) Counter-Baseline (Evaluated on Full 2017 UK Elexon SBP Timeline Across Three Wind Farms).

Operational Wind Farm	**Forecasting / Bidding Strategy**	**Shortfall Penalty (£/(turb·yr))**	**Imbalance Net Cashflow (£/(turb·yr))**	**Pinball Loss** ($q = 1/3$)	**Decision Cost** $\text{Cost}_{\text{static}}$ ($r = 2$)	**Audit Takeaway & Diagnostic Assessment**
Kelmarsh (6 Turbines)	**Zero-Bid** ($q \equiv 0$)	**£0.00**	**+£223,882.14**	0.10143	0.30428	**Degenerate Baseline**: Zero shortfall and apparent cashflow without forecasting
	Persistence Baseline	£7,120.45	+£5,890.12	0.03255	0.09132	Naive operational benchmark
	Unconstrained DLinear	£5,764.03	+£6,584.74	**0.02935**	**0.08804**	Lowest decision risk; cashflow orders of magnitude below Zero-Bid
	Calibrated Distribution Prior	£8,412.28	+£3,485.65	0.03280	0.09841	Shortfall penalty surges; balancing revenue drops
Hill of Towie (21 Turbines)	**Zero-Bid** ($q \equiv 0$)	**£0.00**	**+£239,299.30**	0.10158	0.30473	**Degenerate Baseline**: Apparent cashflow purely from uncontracted physical spill
	Persistence Baseline	£7,450.12	+£9,120.34	0.03241	0.08992	Simple operational benchmark
	Unconstrained DLinear	£5,941.22	+£8,678.09	**0.02971**	**0.08913**	Minimizes genuine non-negative decision risk
	Calibrated Distribution Prior	£5,702.68	+£12,174.95	0.03168	0.09503	Cashflow rises by £3.6k while decision risk degrades +6.5%
Penmanshiel (14 Turbines)	**Zero-Bid** ($q \equiv 0$)	**£0.00**	**+£243,449.80**	0.11230	0.33689	**Degenerate Baseline**: Complete decoupling of cashflow from forecast skill
	Persistence Baseline	£7,080.20	+£6,150.30	0.03433	**0.09586**	Strong linear benchmark in complex terrain
	Unconstrained DLinear	£6,658.86	+£6,377.76	0.03284	0.09852	Sample-restricted unconstrained baseline
	Calibrated Power Curve ($\lambda = 1.0$)	£6,682.66	+£6,382.43	**0.03242**	0.09726	Authentic physical prior successfully reduces decision risk

Notes: (1) Imbalance Cashflow is evaluated as a stylized BSC-price-based imbalance cashflow diagnostic as $\sum(y_t - q_t) \times P_{\text{imb},t}$, where $q_t = \hat{y}_t$ denotes the forward-contracted or nominated energy position in this stylized single-unit accounting model. This stylized diagnostic intentionally abstracts from bilateral contract volume notifications (ECVNs), Balancing Mechanism acceptances, and residual cashflow reallocations. For Zero-Bid ($q \equiv 0$), $y_t - q_t = y_t \geq 0$, resulting in zero shortfall penalty and cashing out all uncontracted energy as physical spill. (2) This analysis serves as an empirical *reductio ad absurdum* demonstrating that standalone unhedged balancing cashflow cannot measure forecasting value without modeling forward commitments ($P_{\text{F}} \times q_t$). (3) Decision Cost ($r = 2$) and Pinball Loss ($q = 1/3$) correctly penalize Zero-Bid for extreme under-commitment ($\text{Cost}_{\text{static}} = \bar{y} \approx 0.30$–$0.34$, matching the mean capacity factor of each site over the test period). (4) DLinear and physical prior rows report multi-seed means across five synchronized random seeds (42, 123, 456, 789, 1024), directly concordant with the canonical benchmarks in Table 2.

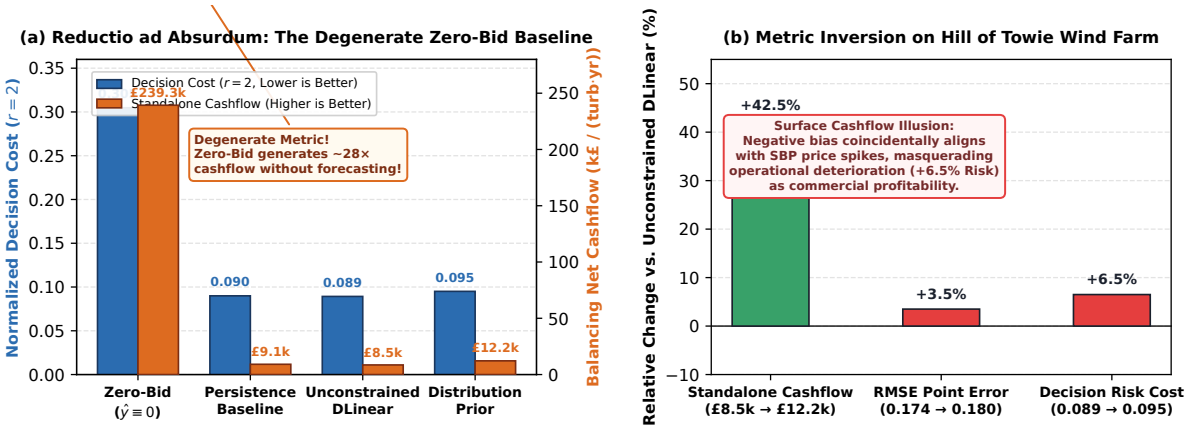


Figure 5: Market Settlement Decoupling and Metric Inversion Diagnostic under Half-Hour Imbalance Accounting. (a) Reductio ad absurdum: the degenerate zero-bid settlement anomaly ($q \equiv 0$) incurs zero shortfall penalty and mechanically generates over £220k/turb·yr in apparent cashflow without generating forecasts, exposing unhedged imbalance cashflow as an evaluation trap. (b) Empirical metric inversion on Hill of Towie, where distribution constraints increase nominal balancing cashflow by +42.5% while concurrently degrading genuine decision risk by +6.5% and RMSE by +3.5%.

Furthermore, on Hill of Towie, distribution regularization increases nominal balancing cashflow by +42.5% (£8,543 to £12,174/(turb·yr); Table 5) while concurrently degrading normalized decision risk ($\text{Cost}_{\text{static}}$) by +6.5% and RMSE by +3.5% (Figure 5b). This divergence occurs because the prior induces a systematic negative bias that happens to coincide with positive price spikes, creating apparent cashflow gains that mask operational dispatch degradation.

7.3. *Dynamic Market Settlement Diagnostic under 2017 UK BSC Prices*

To assess how physical constraints perform under non-stationary market conditions, we evaluated models across full-year 2017 UK Elexon BSC System Buy Prices in a stylized settlement diagnostic (Table 6).

Across dynamic settlement, physical constraints produce divergent operational outcomes across sites: on Kelmarsh, calibrated distribution constraints increase annual shortfall expenditure by £932.56/(turb·yr) (95% CI: [+£648.20, +£1,236.74]) and reduce net balancing cashflow by −£3,264.86/(turb·yr). Conversely, on Hill of Towie, distribution constraints reduce annual shortfall expenditure by £578.57/(turb·yr) and increase net balancing cashflow by £1,481.27/(turb·yr) (95% CI: [+£802.29, +£2,203.32]).

Methodologically, our framework strictly separates two orthogonal sources of empirical variation:

1. **Temporal Autocorrelation Sampling Uncertainty (Delivery Period)**: The bootstrap intervals in Table 6 are estimated via 2,000 replications of 7-day circular block paired bootstrapping (L = 1,008 steps), capturing atmospheric autocorrelation, weather regime shifts, and non-stationary price volatility.
2. **Initialization Stochasticity (Parameter Variance)**: The multi-seed dispersion metrics (mean ± standard deviation) in Table 2 isolate parameter initialization randomness and mini-batch trajectories across identical training data.

7.4. *Supporting Software Forensics: The Physics-Constrained Transformer*

Beyond evaluation metrics, establishing reliable empirical attribution requires verifying that executed software faithfully implements published mathematics. The Physics-Constrained Transformer (PCT [10], *Scientific Reports* 2026; commit `65941ec`) represents the sole publicly accessible training codebase for end-to-end physics-constrained Transformers and serves as an illustrative forensic case study.

In Section 2 of the PCT paper, the authors specify the symmetric Jensen–Shannon (JS) divergence:

$$\mathcal{L}_{\text{JS}}(P_{\text{pred}} \parallel P_{\text{ref}}) = \frac{1}{2}\text{KL}\left(P_{\text{pred}} \middle\| M\right) + \frac{1}{2}\text{KL}\left(P_{\text{ref}} \middle\| M\right), \tag{16}$$

where $M = \frac{1}{2}(P_{\text{pred}} + P_{\text{ref}})$.

However, static AST inspection of the official training script (`train_clean.py`, lines 349–352) reveals that the computational graph evaluates:

$$\mathcal{L}_{\text{code}} = \frac{1}{2}\text{KL}\left(M \parallel P_{\text{pred}}\right) + \frac{1}{2}\text{KL}\left(M \parallel P_{\text{ref}}\right). \tag{17}$$

Table 6: 7-Day Circular Block Paired Bootstrap Contrasts vs. Unconstrained DLinear under Calibrated Policy (2,000 Replications).

Site	Method Contrast	Dynamic Cost Diff $\Delta\text{Cost}_{\text{dyn}}$ (95% CI)	Shortfall Cashout Diff £/(turb·yr) (95% CI)	Balancing Cashflow Diff £/(turb·yr) (95% CI)
Hill of Towie	Bounds vs. DLinear	+0.000000 ([+0.000000, +0.000000])	+£0.15 ([−0.17, +0.51])	−£0.32 ([−1.37, +0.62])
	Power Curve vs. DLinear	−0.000008 ([−0.000021, +0.000007])	+£22.99 ([+10.93, +38.43])	−£89.40 ([−133.51, −54.70])
	Distribution Prior vs. DLinear	+0.000766 ([+0.000277, +0.001201])	−£578.57 ([−834.86, −367.51])	+£1,481.27 ([+802.29, +2,203.32])
	Ridge vs. DLinear	−0.000733 ([−0.001315, −0.000125])	+£192.50 ([−26.66, +445.74])	−£1,151.96 ([−1,567.98, −795.28])
Kelmarsh	Bounds vs. DLinear	+0.000001 ([+0.000000, +0.000002])	+£1.72 ([−0.01, +4.63])	−£4.62 ([−12.67, +0.05])
	Power Curve vs. DLinear	−0.000039 ([−0.000054, −0.000023])	−£16.61 ([−33.12, +0.87])	+£19.51 ([−27.76, +62.86])
	Distribution Prior vs. DLinear	+0.000932 ([+0.000423, +0.001385])	+£932.56 ([+648.20, +1,236.74])	−£3,264.86 ([−4,365.40, −2,223.50])
	Ridge vs. DLinear	+0.001338 ([+0.000917, +0.001794])	+£750.61 ([+330.70, +1,151.06])	−£690.94 ([−1,669.76, +409.51])

Interpretation: Negative shortfall cashout difference indicates cost reduction relative to unconstrained DLinear. Positive balancing cashflow difference indicates higher net cashflow component credited from the balancing mechanism. Confidence intervals estimated via 2,000 replications of 7-day circular block paired bootstrapping to account for temporal autocorrelation.
Site Evaluation Scope: Kelmarsh and Hill of Towie feature continuous, uninterrupted full-calendar 2017 SCADA records enabling continuous 7-day circular block bootstrap resampling. Penmanshiel serves as a sample-restricted stress-testing benchmark (N_{train} = 1,583, Table 2); because its evaluated records lack unbroken full-calendar continuity, it is evaluated under multi-seed static dispatch ablations rather than calendar-level 7-day circular block dynamic settlement.

This occurs because PyTorch's `torch.nn.functional.kl_div(input, target)` computes $\sum_i \text{target}_i(\log \text{target}_i - \text{input}_i) = \text{KL}(\text{target} \,\|\, \text{input})$. Passing `pred_kde.log()` as input binds P_{pred} to the denominator, executing what we designate the **reverse-argument JS form**.

Differentiating $\mathcal{L}_{\text{code}}$ directly with respect to $P_{\text{pred},i}$ yields:

$$\frac{\partial \mathcal{L}_{\text{code}}}{\partial P_{\text{pred},i}} = \frac{1}{4}\left[\log\left(\frac{M_i^2}{P_{\text{pred},i}P_{\text{ref},i}}\right) + 1 - \frac{P_{\text{ref},i}}{P_{\text{pred},i}}\right]. \quad (18)$$

Equation (18) exposes an analytical **inverse-density singularity**: whenever $P_{\text{pred},i} \to 0$ in a region where empirical data exists ($P_{\text{ref},i} > 0$), the ratio $\frac{P_{\text{ref},i}}{P_{\text{pred},i}}$ diverges as $O(1/\epsilon)$. In contrast, for the mathematically specified Standard-JS in Eq. (16), the partial derivative is bounded above by $\frac{1}{2}\ln 2 \approx 0.3466$ and exhibits only mild logarithmic divergence ($O(|\log \epsilon|)$).

Empirical Gradient and Dispatch Contrast. Empirical training on Kelmarsh and Penmanshiel across five seeds confirms that this implementation divergence materially alters optimization (Figure 6). At $\lambda = 1.0$, the reverse-argument JS form amplifies mean gradient L_2 norms by 2.20× on Kelmarsh (0.1665 ± 0.0160 vs. 0.0756 ± 0.0048) and by 2.72× on Penmanshiel (0.2465 ± 0.0217 vs. 0.0907 ± 0.0067; Figure 6a). At weak regularization ($\lambda = 0.01$ on Penmanshiel), Corrected Standard-JS achieves a dispatch cost of 0.09804 ± 0.00024, whereas the official reverse-argument implementation inflates dispatch cost to 0.09997 ± 0.00086 (a +1.35% degradation; Figure 6b). Under strong regularization ($\lambda = 1.0$), excessive penalization degrades operational dispatch across both distribution prior variants.

Negative Control: Algebraic Verification of 2D Product-Kernel KDE. To establish an objective negative control and verify that our audit does not falsely report errors where none exist, we algebraically verified the tensor outer product $G = K_p^\top K_w \in \mathbb{R}^{B_p \times B_w}$ in the PCT codebase. Expanding the matrix product element at grid coordinate (i, j) yields:

$$\left[K_p^\top K_w\right]_{i,j} = \sum_{s=1}^{N} \phi\left(\frac{b_i^p - P_s}{h_p}\right) \cdot \phi\left(\frac{b_j^w - w_s}{h_w}\right). \quad (19)$$

Equation (19) is the exact analytical definition of a standard **paired 2D product-kernel Kernel Density Estimator** on correlated bivariate data. Because the summation contracts over concurrent sample index s, empirical joint coupling between wind speed and power is preserved. This confirms that software discrepancies in the audited codebase were strictly localized to the divergence argument order, rather than upstream kernel estimation.

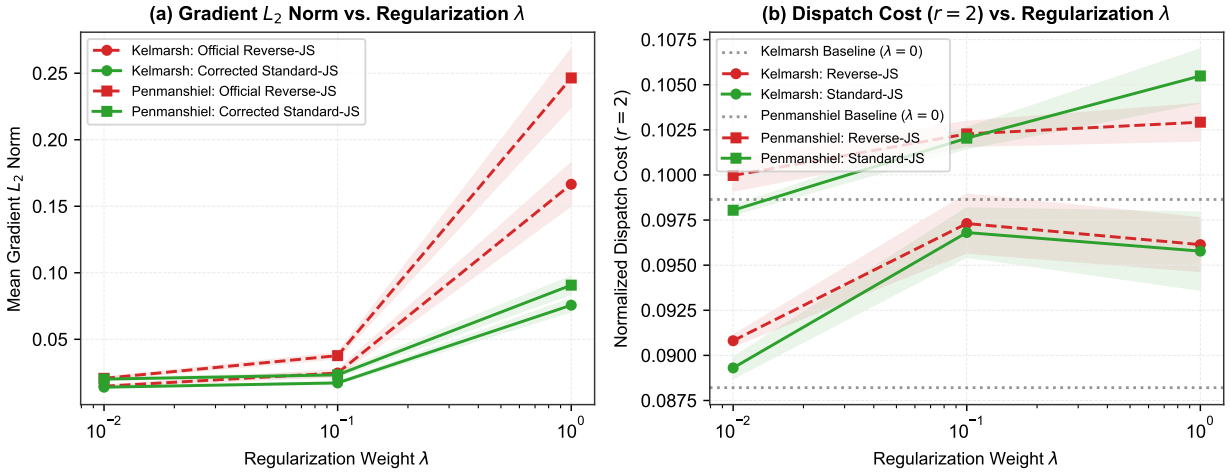


Figure 6: Empirical Optimization Contrast between Official Reverse-JS and Corrected Standard-JS on Kelmarsh and Penmanshiel. (a) Mean gradient L_2 norm: the reverse-argument JS form induces substantial gradient amplification at $\lambda = 1.0$ (2.20× on Kelmarsh, 2.72× on Penmanshiel) due to the $-\frac{P_{\text{ref},i}}{4P_{\text{pred},i}}$ singularity ($O(1/\epsilon)$ in Eq. (18)). (b) Operational dispatch cost ($r = 2$): Corrected Standard-JS prevents premature performance degradation at weak $\lambda = 0.01$, while excessive regularization degrades dispatch across both distribution prior variants.

8. Discussion, Conceptual Synthesis, and Reporting Governance

In this section, we synthesize our empirical and theoretical findings into a conceptual bias–variance framework, explicitly delineate what our controlled experiments formally identify versus what remains unmeasured, document falsified initial conjectures, and formulate an actionable operational governance protocol.

8.1. Conceptual Bias–Variance Synthesis: Understanding the Operating Envelope

The empirical patterns mapped across our benchmarks can be conceptually interpreted through the trade-off between variance reduction and structural approximation bias, serving strictly as conceptual interpretations consistent with—not identified by—the observed learning curves:

1. **Tested Sample-Restricted Regimes** ($N_{\text{train}} \in \{500, 1{,}000\}$): Under a conceptual variance-reduction interpretation, data-driven forecasters in sample-restricted regimes are hypothesized to suffer from elevated parameter estimation variance, rendering them prone to non-physical overfitting. Under strong regularization ($\lambda = 1.0$), an authentic empirical power curve constrains the hypothesis space to meteorologically plausible manifolds, providing substantial downside risk reduction (−1.28% on Penmanshiel, −2.00% on Kelmarsh, −3.62% on Hill of Towie). However, if the prior is misspecified or permuted, this rigid constraint enforces counter-physical parameter updates, triggering severe performance degradation (+61.2%).

2. **Largest Evaluated Training Budget** ($N_{\text{train}} = 5{,}000$): Purely data-driven representations can fit the observed input–output relationship more flexibly as training coverage expands from operational telemetry. Conceptually, any variance-reduction benefit offered by a fixed prior is hypothesized to diminish relative to emerging structural approximation bias inherent in a static, one-dimensional empirical curve. At the largest evaluated training budget ($N_{\text{train}} = 5{,}000$), unconstrained DLinear matches or slightly outperforms the fixed prior (+0.33% on Kelmarsh, +1.73% on Hill of Towie).

3. **Penalty Scaling and Numerical Smoothing**: Under weak regularization ($\lambda = 0.01$), genuine physical priors and scrambled pseudo-priors yield closely matched performance. This lack of empirical separation demonstrates that weak constraints function primarily as non-specific numerical regularizers, challenging the assumption that marginal gains under small penalty weights represent physical understanding.

We emphasize that this bias–variance perspective serves strictly as a **conceptual interpretation** consistent with—not identified or proven by—the observed learning curves, rather than an empirically measured algebraic

identity. Taken together, our learning-curve and penalty-sensitivity experiments suggest that both data availability and regularization strength jointly govern the operating envelope of physical priors.

8.2. Attributional Scope: What Is Evaluated vs. What Is Not

To preserve methodological clarity, we explicitly distinguish the attributional scope established by our experiments from unresolved confounding factors:

Empirically Distinguished Under the Controlled Protocol.

(a) **Domain Specificity under Strong Regularization**: Under the tested matched-control protocol, divergence between authentic and pseudo-scrambled priors provides evidence that the observed performance shift is specific to the empirical aerodynamic mapping, rather than being explained by penalty form and weight alone.

(b) **Decoupling from Historical Information Budgets**: The strictly budget-matched information ablation ($N_{\text{prior}} = N_{\text{train}} = 500$) confirms that domain-structured prior effects retain 77% to 92% of the full-history mean benefit, demonstrating that the majority of the observed gain cannot be explained by auxiliary historical data volume alone.

(c) **Attenuating Returns across Training Volumes**: Subsampled within-site learning curves show that prior advantages diminish as training volume expands, reversing slightly under the evaluated strong-prior setting at $N_{\text{train}} = 5{,}000$.

(d) **Vulnerability to Prior Distortion and Latency**: Continuous corruption scans identify an empirical crossover region around $\alpha \approx 0.07$ on Penmanshiel, while the Hill of Towie lag scan reveals that decision risk exceeds the unconstrained reference by the first evaluated lag $k = 6$ (1 h delay) and deteriorates monotonically through $k = 36$ when training losses rely on stale historical weather proxies. These values are empirical crossover locations along the evaluated perturbation grid, not population-level physical thresholds.

(e) **Prediction–Decision Decoupling**: The Info-Symmetric benchmark confirms that reducing RMSE can simultaneously worsen asymmetric dispatch loss, and the Zero-Bid reductio ad absurdum proves that standalone unhedged balancing cashflow is structurally decoupled from forecasting skill.

(f) **Software Discrepancy in Divergence Losses**: Static AST analysis and gradient derivations identify an analytical $O(1/\epsilon)$ gradient singularity in the open-source PCT codebase, confirming that implementation divergence can alter optimization trajectories.

What Is Not Identified (Confounding Disclaimers).

(i) **Cross-Site Topographical Causation**: Differences across Kelmarsh, Hill of Towie, and Penmanshiel confound topographic complexity with turbine ratings (2.05 MW vs. 2.30 MW), hub heights, wake geometry, and local micro-climates. We do not assert that complex terrain causally mandates physical constraints; topography remains an observational covariate.

(ii) **Universal Causal Effect of Physics**: Our experiments evaluate specific empirical power-curve and distributional constraints under evaluated configurations; they do not establish universal causal claims for "physics" as an abstract treatment across all energy systems.

(iii) **Global Generalization of Auditing Metrics**: The 95.7% code deficit and 50.0% confounded ablation rates characterize the prespecified 46-study scoping cohort (2023–2026), rather than all applied AI literature. Similarly, the absence of documented failure regimes describes the reporting scope of the surveyed publications (which reported exclusively within nominal operating envelopes) rather than identifying underlying researcher behavior or file-drawer effects.

(iv) **Universal Software Defects**: Demonstrating an implementation divergence in PCT confirms that code-manuscript mismatches occur, but does not imply that all physics-informed forecasting codebases contain errors.

(v) **Universality of Decoupled Calibration**: Counterexamples 1 and 2 are formal existence proofs establishing theoretical limits under spatial heteroscedasticity, not an empirical claim that scalar calibration universally underperforms joint training.

8.3. Two Conjectures Falsified During the Audit

Documenting falsified initial hypotheses provides an auditable record of our inquiry:

1. *Bivariate Coupling in Tensor Outer Products*: We initially hypothesized that the tensor product $K_p^\top K_w$ in PCT assumed statistical independence ($p(w, P) = p(w)p(P)$). Formal algebraic expansion (Equation (19)) refuted this conjecture, proving that contracting over batch samples preserves authentic bivariate product-kernel KDE coupling.

2. *Price Asymmetry Dominance over Prior Regularization*: We initially conjectured that physical distribution priors would consistently improve balancing market cashflows across all sites. Dynamic settlement analysis refuted this hypothesis: distribution priors reduced shortfall costs on Hill of Towie while increasing shortfall penalties on Kelmarsh, demonstrating that price spike timing dominates physical regularizers in unhedged cashflow calculations.

8.4. Actionable Guidance: The Five-Step Practical Decision Rule

Before deploying physical priors into operational forecasting systems, engineers and system operators should evaluate five sequential criteria:

1. **Prior Specificity Check**: Does the authentic physical prior outperform a form- and weight-matched scrambled negative control under the planned regularization weight? If genuine and scrambled priors yield closely matched outcomes, the observed gain should be treated as non-specific numerical smoothing.

2. **Information Budget Alignment Check**: When fitted strictly on the available training sample window without auxiliary historical archives, does the prior retain the majority of its performance advantage? If not, the apparent gain is attributable to auxiliary historical data volume rather than inductive physical guidance.

3. **Sample Volume and Regularization Annealing Check**: As training data expands, is the regularization weight λ systematically annealed? In well-sampled regimes, rigid constraints should be relaxed to mitigate potential structural approximation bias from degrading predictions.

4. **State Alignment and Latency Tolerance Check**: Does the physical prior rely on lagged meteorological proxies? In operational deployment, stress tests must verify whether the prior's decision advantage survives expected telemetry and NWP delivery latencies.

5. **Decision-Aligned Metric Check**: Is the prior evaluated against asymmetric operational dispatch risk ($\mathcal{L}_{\text{disp}}$ at the operational penalty ratio) rather than symmetric point errors (RMSE) or unhedged balancing cashflows? Models that reduce RMSE while elevating shortfall risk should be barred from deployment.

8.5. Methodological Verification Checklist (Criteria C1–C5)

Table 7 synthesizes these findings into five verification criteria (C1–C5), adapting the reporting principles of the REFORMS consensus recommendations [30] and DOME guidelines [31] to physics-constrained machine learning.

Adopting these five criteria will ensure that future physics-constrained renewable energy forecasting studies provide transparent, reproducible, and decision-aligned empirical evidence.

9. Conclusion

Three overarching conclusions emerge from this empirical audit regarding physics-constrained wind power forecasting.

First, at the literature and reporting level, studies in the audited scoping cohort reported exclusively within nominal operating envelopes (0/46 documenting failure regimes), while frequently relying on confounded module ablations and omitting public code. When computational graphs are subjected to static and gradient verification, implementations can diverge substantially from published equations. Establishing reliable scientific claims in physics-constrained machine learning requires open source code, computational graph verification, and form-matched negative controls.

Second, at the methodological and inductive level, physical priors function not as universal boosters, but as **conditional inductive guardrails**. In restricted training windows, authentic aerodynamic curves supply vital inductive regularization that curtails tail dispatch risk, exhibiting prior-specific mapping effects in this sample-restricted setting that cannot be attributed to data volume alone. As operational telemetry expands, purely data-driven models match or slightly exceed fixed priors—a dynamic interpreted through the lens of diminishing variance reduction and emerging structural approximation bias, though these mechanisms remain conceptual interpretations consistent with—not identified by—the observed learning curves. Furthermore, physical priors exhibit pronounced fragility under uncalibrated engineering drift and stale telemetry proxies (where decision risk exceeds the unconstrained reference by the first evaluated lag $k = 6$ / 1 h delay and deteriorates monotonically

Table 7: Methodological Verification Checklist for Physics-Constrained Wind Power Forecasting (Criteria C1–C5).

Criterion	Governance Dimension	Mandatory Audit Protocol	Common Pitfall / Anti-Pattern	Required Verification Artifact
C1	**Isolated Prior Ablation**	Train identical network backbones with constraint penalty $\lambda = 0$ vs. $\lambda > 0$ under identical seed schedules.	Confounding module excision (removing entire GNNs or attention blocks) with physical knowledge.	Table of $\lambda = 0$ vs. $\lambda > 0$ with identical parameter counts and multi-seed variances.
C2	**Domain Specificity Control**	Evaluate pseudo-scrambled priors that preserve loss formulation, architecture, and regularization weight λ while permuting physical relations.	Mistaking non-specific numerical regularization under weak penalty settings (e.g., $\lambda = 0.01$ under the present loss scaling) for domain reasoning.	Scrambled control performance curve and empirical specificity ratio (p-value contrast).
C3	**Proxy Staleness Stress-Test**	If constraints depend on physical inputs (e.g., wind speed), scan historical proxy lag $k \in \{6, 12, 18, 36\}$ steps to evaluate proxy staleness when forward NWP forecasts are absent.	Assuming temporally aligned physical proxies when constructing the offline constraint target.	Observation staleness degradation curve demonstrating operational harm horizon.
C4	**Metric Decoupling & Accounting**	Report symmetric error (RMSE), non-negative asymmetric decision risk (Pinball), and test degenerate zero-bid anomaly baselines.	Reporting unhedged standalone imbalance cashflow in isolation, mistaking accidental price-spike alignment for skill.	Decoupled market evaluation coupling forward commitments and balancing cashflow.
C5	**Computational Graph Parity**	Verify that software loss implementations match published mathematical equations in argument direction and normalization.	Argument order flip in asymmetric divergences (e.g., PyTorch `F.kl_div`) causing reverse optimization.	Static code audit report and unit test verifying loss tensor gradient directions.

Notes: Derived from our empirical audit of 46 scoping studies and adapted from general machine-learning reporting principles [30, 31]. We propose C1–C5 as an operational reporting checklist for future physics-constrained renewable energy forecasting studies.

through $k = 36$). Operational deployment therefore mandates adaptive regularization annealing and telemetry latency tolerance checks.

Third, at the operational decision level, symmetric regression metrics (RMSE and MAE) and standalone unhedged balancing cashflows structurally decouple from real-world dispatch risk. Symmetric metrics mask catastrophic tail under-predictions during shortfall events, while unhedged single-component settlement rewards degenerate, non-forecasting strategies. Forecasting models intended for grid integration must be optimized and audited against decision-aligned asymmetric dispatch objectives.

In conclusion, physics-constrained machine learning for energy systems must move beyond uncritical performance narratives. By institutionalizing form-matched negative controls, boundary stress tests, computational graph auditing, and decision-aligned evaluation (Criteria C1–C5), the field can transition toward verifiable, reproducible, and operationally dependable forecasting systems.

Appendix A. Mathematical Proofs of Boundary Counterexamples

This appendix section provides formal mathematical proofs for the two boundary counterexamples presented in Section 3, delineating the theoretical limits of decoupled decision calibration.

Appendix A.1. Proof of Counterexample 1: Constant Shift Invariance and Zero Excess Risk

Counterexample 1 (Constant Shift Invariance). Let the target variable follow a uniform distribution $Y \sim U[0.25, 0.75]$, and consider an asymmetric cost ratio $r = 2$ ($\tau = 1/3$). Suppose an unconstrained model outputs the exact conditional expectation $f_0 = \mathbb{E}[Y] = 0.50$, while a physics-constrained model incurs a constant positive shrinkage shift $c = +0.05$, outputting $f_\lambda = 0.55$.

Prediction Representation Error (MSE). The mean squared prediction errors are:

$$\mathrm{MSE}(f_0) = \mathrm{Var}(Y) = \frac{0.50^2}{12} \approx 0.020833, \tag{A.1}$$

$$\mathrm{MSE}(f_\lambda) = \mathrm{MSE}(f_0) + c^2 \approx 0.023333. \tag{A.2}$$

The physical constraint strictly inflates prediction MSE by $\Delta\mathrm{MSE} = +0.0025$ (+12.0%).

Uncalibrated Operational Dispatch Loss. Under uncalibrated dispatch ($p = f$), the expected asymmetric loss is:

$$\begin{aligned}\mathcal{R}_{\text{uncal}}(p) &= 2\int_{0.25}^{p}(p-y)\cdot 2\,dy + \int_{p}^{0.75}(y-p)\cdot 2\,dy \\ &= 2(p-0.25)^2 + (0.75-p)^2.\end{aligned} \tag{A.3}$$

Evaluating at each prediction:

$$\mathcal{R}_{\text{uncal}}(f_0 = 0.50) = 2(0.25)^2 + (0.25)^2 = 0.187500, \tag{A.4}$$

$$\mathcal{R}_{\text{uncal}}(f_\lambda = 0.55) = 2(0.30)^2 + (0.20)^2 = 0.220000. \tag{A.5}$$

Post-Hoc Decoupled Quantile Calibration. The theoretical Bayes-optimal quantile at $\tau = 1/3$ is:

$$q_{1/3} = 0.25 + \tfrac{1}{3}(0.75 - 0.25) \approx 0.416667. \tag{A.6}$$

Under decoupled quantile calibration, each model searches for its optimal scalar displacement:

$$\delta_0^* = q_{1/3} - f_0 \approx 0.416667 - 0.500000 = -0.083333, \tag{A.7}$$

$$\delta_\lambda^* = q_{1/3} - f_\lambda \approx 0.416667 - 0.550000 = -0.133333. \tag{A.8}$$

Notice that the resulting dispatch commitments are identical:

$$p_0^* = f_0 + \delta_0^* = 0.416667, \quad p_\lambda^* = f_\lambda + \delta_\lambda^* = 0.416667. \tag{A.9}$$

Consequently, the post-calibration operational risks are identical:

$$\mathcal{R}_{\text{cal}}(p_0^*) = \mathcal{R}_{\text{cal}}(p_\lambda^*) = 0.166667 \implies \Delta\mathcal{R}_{\text{disp}} \equiv 0. \tag{A.10}$$

Theoretical Takeaway. Counterexample 1 proves that **an increase in prediction MSE is compatible with invariant post-calibration operational risk**. When constraint-induced representation error consists of a uniform shift, scalar calibration absorbs the distortion entirely, establishing that strict positive excess risk lower bounds cannot hold unconditionally.

Appendix A.2. Proof of Counterexample 2: Spatial Heteroscedasticity and Risk Reversal

Counterexample 2 (Spatial Heteroscedasticity). Let $X \in \{0, 1\}$ be a binary indicator with $P(X = 0) = P(X = 1) = 0.5$. Let the conditional target distributions exhibit heteroscedastic spread:

$$Y \mid (X = 0) \sim U[0.20, 0.60] \quad (\mu_0 = 0.40, s_0 = 0.40), \tag{A.11}$$

$$Y \mid (X = 1) \sim U[0.30, 0.90] \quad (\mu_1 = 0.60, s_1 = 0.60). \tag{A.12}$$

Let the asymmetric cost ratio be $r = 2$ ($\tau = 1/3$). Consider two candidate forecasters: Model A (the unbiased conditional mean) with predictions $f_A(0) = 0.40$ and $f_A(1) = 0.60$; and Model B (a biased estimator with higher MSE) with predictions $f_B(0) = 0.40$ and $f_B(1) = 0.55$.

Prediction MSE Comparison. The mean squared prediction errors are:

$$\begin{aligned}\text{MSE}(f_A) &= 0.5\left(\frac{0.40^2}{12}\right) + 0.5\left(\frac{0.60^2}{12}\right) \\ &= \frac{13}{600} \approx 0.021667,\end{aligned} \tag{A.13}$$

$$\begin{aligned}\text{MSE}(f_B) &= \text{MSE}(f_A) + 0.5(0.55 - 0.60)^2 \\ &= 0.021667 + 0.001250 \approx 0.022917.\end{aligned} \tag{A.14}$$

Model B has strictly higher prediction MSE by $\Delta\text{MSE} = +0.001250$ (+5.8%).

Post-Hoc Global Scalar Calibration. The local Bayes-optimal quantiles at $\tau = 1/3$ are:

$$q_{1/3}(0) = 0.20 + \tfrac{0.40}{3} = \tfrac{1}{3} \approx 0.333333, \tag{A.15}$$

$$q_{1/3}(1) = 0.30 + \tfrac{0.60}{3} = 0.500000. \tag{A.16}$$

The Bayes-optimal operational dispatch loss is $\mathcal{R}(q_\tau) = 0.5(\frac{0.40}{3}) + 0.5(\frac{0.60}{3}) = \frac{1}{6} \approx 0.166667$. The local conditional densities are $f_0(y) = 2.5$ and $f_1(y) = 1.6667$.

Because local quantile-to-mean shifts differ ($\Delta_0 = q_{1/3}(0) - \mu_0 = -\frac{1}{15} \approx -0.066667$ versus $\Delta_1 = q_{1/3}(1) - \mu_1 = -0.100000$), a single global scalar offset δ^* cannot simultaneously achieve both local quantiles. The expected operational risk under global shift δ is:

$$\mathcal{R}(f + \delta) = 0.5\,\mathcal{R}_0(f(0) + \delta) + 0.5\,\mathcal{R}_1(f(1) + \delta). \tag{A.17}$$

For **Model A** ($f_A = \mu$), the first-order optimality condition for δ_A^* is:

$$(r + 1)\left[0.5 \cdot 2.5(\delta_A^* - \Delta_0) + 0.5 \cdot 1.6667(\delta_A^* - \Delta_1)\right] = 0, \tag{A.18}$$

yielding the density-weighted optimal global offset:

$$\delta_A^* = \frac{2.5(-\frac{1}{15}) + 1.6667(-0.10)}{2.5 + 1.6667} = -0.080000. \tag{A.19}$$

The excess operational risk for Model A evaluates to:

$$\begin{aligned}\mathcal{R}(f_A + \delta_A^*) - \mathcal{R}(q_\tau) &= \frac{3}{4} \cdot 2.5(-0.0800 + 0.06667)^2 \\ &\quad + \frac{3}{4} \cdot 1.6667(-0.0800 + 0.1000)^2 \\ &= 0.000833.\end{aligned} \tag{A.20}$$

For **Model B** ($f_B(0) = 0.40$, $f_B(1) = 0.55$), the quantile shifts relative to Model B predictions are:

$$\Delta_{B,0} = q_{1/3}(0) - f_B(0) = -0.066667, \tag{A.21}$$

$$\Delta_{B,1} = q_{1/3}(1) - f_B(1) = -0.050000. \tag{A.22}$$

Because Model B's negative bias at $X = 1$ brings its prediction closer to $q_{1/3}(1)$, the spread between local shifts shrinks from 0.0333 to 0.0167. The optimal global offset for Model B is:

$$\begin{aligned}\delta_B^* &= \frac{2.5(-0.066667) + 1.6667(-0.050000)}{2.5 + 1.6667} \\ &= -0.060000.\end{aligned} \tag{A.23}$$

Evaluating the excess operational risk for Model B:

$$\begin{aligned}\mathcal{R}(f_B + \delta_B^*) - \mathcal{R}(q_\tau) &= \frac{3}{4} \cdot 2.5(-0.0600 + 0.06667)^2 \\ &\quad + \frac{3}{4} \cdot 1.6667(-0.0600 + 0.0500)^2 \\ &= 0.000208.\end{aligned} \tag{A.24}$$

Theoretical Takeaway. Comparing excess operational risks reveals:

$$\begin{aligned}\mathcal{R}(f_B + \delta_B^*) - \mathcal{R}(q_\tau) &= 0.000208 \\ &< 0.000833 = \mathcal{R}(f_A + \delta_A^*) - \mathcal{R}(q_\tau).\end{aligned} \tag{A.25}$$

Model B achieves a **75% reduction in excess operational risk** despite having +5.8% higher prediction MSE. Counterexample 2 proves that under spatial heteroscedasticity, an estimator with higher MSE can achieve strictly lower post-calibration dispatch risk than the true conditional expectation.

Appendix B. Systematic Literature Scoping Protocol (PRISMA-ScR)

We conducted our empirical scoping audit in accordance with the PRISMA Extension for Scoping Reviews (PRISMA-ScR) guidelines [32]. Literature retrieval was executed across two complementary channels covering January 1, 2023, through March 1, 2026, with rolling monitoring through July 2026. First, programmatic queries via the OpenAlex API using four targeted title queries (Q_1: `"physics-constrained" AND "wind"`, Q_2: `"physics-informed" AND "wind power"`, Q_3: `"physics-guided" AND "wind"`, Q_4: `"physics-embedded" AND "wind power"`) identified 72 candidate records. Second, targeted searches on Google Scholar, arXiv, and preprint servers yielded 17 additional records ($N = 89$ total).

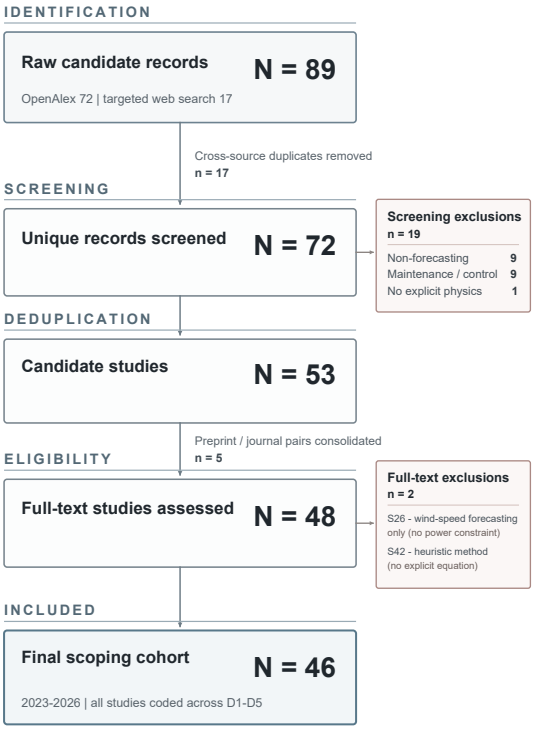


Figure B.7: PRISMA-ScR Flow Diagram Illustrating Scoping Cohort Derivation (89 → 72 → 53 → 48 → 46). Cross-source deduplication eliminated 17 duplicates; title/abstract screening excluded 19 records; study-level deduplication consolidated 5 preprint/journal pairs; and full-text eligibility screening excluded 2 studies, leaving 46 eligible full-text studies.

Screening followed structured multi-phase filtering (Figure B.7): cross-source deduplication eliminated 17 duplicate records, leaving 72 unique entries. Title and abstract screening excluded 19 records (non-forecasting structural health monitoring, active pitch control, or purely data-driven studies). Study-level deduplication consolidated five preprint/journal pairs sharing identical authorship, datasets, and equations, reducing candidate entries from 53 to 48. Full-text evaluation against explicit eligibility criteria excluded two studies (S26 forecasted wind speed exclusively without a power constraint;

S42 implemented heuristic feature clustering without physical equations), establishing our final representative scoping corpus of $N = 46$ **unique scoping studies** (31 journal records, 4 conference records, 10 records with SSRN/arXiv/working-paper identifiers, and 1 record without a DOI). A supplementary sensitivity query expanding retrieval across Scopus and IEEE Xplore identified 14 additional papers, with full-text inspection supporting the same reporting-gap pattern; these records are treated as a sensitivity check rather than pooled into the 46-study denominator.

Appendix C. Multi-Rater Reliability Benchmarking

To assess the empirical stability of the taxonomy before full extraction, three raters (Coder A, Coder B, and Coder C) independently conducted a rigorous, fully blind pilot inter-rater reliability test on a stratified representative subset of 20 candidate studies across all five dimensions. Study eligibility screening achieved unanimous consensus (Fleiss $\kappa = 1.000$). Parameter provenance ($\mathcal{D}_1$) demonstrated perfect agreement ($\kappa = 1.000$), while application stage ($\mathcal{D}_2$) yielded near-perfect agreement ($\kappa = 0.950$ overall, $\kappa = 0.938$ on included studies). Inference telemetry requirements ($\mathcal{D}_3$) achieved substantial agreement ($\kappa = 0.743$ overall, $\kappa = 0.646$ on included studies).

Implementation fidelity ($\mathcal{D}_4$) exhibited moderate concordance prior to consensus arbitration ($\kappa = 0.716$ overall in the stratified pilot, but $\kappa = 0.341$ when evaluated strictly on included studies). This reduction reflects the classical prevalence paradox described by Feinstein and Cicchetti: because 95.7% of candidate studies omitted public code, expected chance agreement was high, which depresses Fleiss' κ despite a raw observed agreement of 90.7%. For ablation semantics ($\mathcal{D}_5$), raters achieved strong concordance ($\kappa = 0.851$ overall, $\kappa = 0.818$ on included studies). On harm reporting, all three raters unanimously coded zero instances of harm documentation across the included pilot studies (0/18); because marginal variance was zero, Cohen's κ is mathematically indeterminate and is formally reported as 100% observed agreement ($P_o = 1.000$).

Following pilot protocol stabilization and consensus arbitration, full extraction across all 46 included studies was conducted with complete traceability: rather than relying on ungrounded categorical assignments, every coding decision across all five dimensions ($\mathcal{D}_1$–$\mathcal{D}_5$) is anchored in the open extraction database and supplementary material by an auditable verbatim quote evidence trail, citing exact section numbers, equation indices, table identifiers, and verbatim textual excerpts (Table B.8, Table B.9).

Declaration of Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Author Contributions (CRediT)

Junyu Li: Conceptualization, Methodology, Software, Formal analysis, Data curation, Investigation, Visualization, Project administration, Writing – original draft, Writing – review & editing. **Juntao Du**: Methodology (review protocol), Validation, Data curation (scoping extraction double-coding), Writing – review & editing.

Declaration of Generative AI in the Writing Process

During the preparation of this work, the authors used ChatGPT and Google Gemini to assist with literature synthesis, manuscript organization, language refinement, code review, and consistency checking. All scientific claims, calculations, references, experimental results, and final wording were independently reviewed and verified by the authors, who take full responsibility for the content.

Acknowledgments

We thank the open science teams at Zenodo, Cubico Sustainable Investments, RES, TRIG, and Elexon for maintaining transparent public repositories of operational renewable energy telemetry and market clearing records. Computations were supported by dedicated high-performance computing infrastructure.

Data and Code Availability

Raw operational SCADA telemetry from the Kelmarsh (DOIs: 10.5281/zenodo.5841834) and Penmanshiel (10.5281/zenodo.5946808) wind farms is openly available on Zenodo under CC-BY 4.0 licenses from Cubico Sustainable Investments. Operational SCADA telemetry from the Hill of Towie wind farm is openly available on Zenodo under CC-BY 4.0 license released by RES on behalf of The Renewables Infrastructure Group (TRIG) (DOI: 10.5281/zenodo.20204946). Elexon System Buy Price records are freely accessible

Table B.8: Empirical Distribution of Constraint Lifecycle Dimensions ($\mathcal{D}_1$–$\mathcal{D}_5$) Across 46 Audited Scoping Studies (2023–2026).

Lifecycle Dimension	Taxonomic Category	Count ($N = 46$)	Proportion (%)	Typical Exemplars in Audited Literature
D1: Parameter Provenance	`intrinsic`	40	87.0%	Betz aerodynamic limit ($C_p \leq 16/27$), rated capacity P_{rated}, Jensen wake deficit
	`historical-static`	5	10.9%	Empirical bivariate power curve ($P = f(v)$), 2D KDE distribution estimated offline on train SCADA
	`learned-online`	1	2.2%	Auxiliary sub-network dynamically predicting time-varying aerodynamic coefficients $C_p(t)$
D2: Application Stage	`training-loss`	21	45.7%	Differentiable penalty term in backpropagation ($\mathcal{L} = \mathcal{L}_{\text{MSE}} + \lambda\mathcal{L}_{\text{phys}}$)
	`inference-arch.`	23	50.0%	Wake-guided GNN adjacency matrix, monotonic activation layers, dual-stream physical fusion
	`data-generation`	1	2.2%	Physics-guided generative diffusion model synthesizing extreme wind generation scenarios
	`post-hoc-correction`	1	2.2%	Post-inference conformal prediction and residual projection layers
D3: Inference Requirement	`none`	36	78.3%	Standard historical autoregressive SCADA sequence; zero runtime external dependencies
	`forecast-input`	6	13.0%	Requires contemporaneous/future NWP meteorological forecasts over $[t+1, t+H]$
	`contemp.-state`	4	8.7%	Requires synchronized real-time operational SCADA from adjacent/upstream wind turbines
D4: Implementation Fidelity	`not-found`	44	95.7%	No accessible public repository or open-source software identified under recorded protocol
	`code-mismatch`	1	2.2%	PCT: loss function calculates reverse-argument JS form rather than specified standard JS
	`code-aligned`	1	2.2%	PINN-UQ: algebraic electromechanical tie constraints faithfully matching published equations
D5: Evaluation Validity & Harm Reporting	`module-ablation`	23	50.0%	Excises multi-layer neural submodule; conflates parametric capacity with physical bias
	`constraint-on-off`	17	37.0%	Toggles penalty multiplier ($\lambda = 0$ vs. $\lambda > 0$) under strictly identical network capacity
	`both`	6	13.0%	Implements both parameter toggles and architectural ablations across comparative baselines

Table B.9: Semantic Classification and Attributional Interpretability of Reported Ablation Studies Across the Audited Scoping Corpus ($N = 46$).

Reported Ablation Semantic	Count ($N = 46$)	Proportion (%)	Attributional Validity and Interpretability
`constraint-on-off`	17	37.0%	**Strict Isolation**: Explicitly evaluates $\lambda = 0$ vs. $\lambda > 0$ under identical neural architecture, isolating the physical regularizer under matched parameter capacity.
`both`	6	13.0%	**Strict Isolation**: Executes both isolated parameter toggles ($\lambda = 0$) and broader structural ablations, satisfying unconfounded control.
`module-ablation`	23	50.0%	**Broader Architecture**: Excises an entire neural submodule (e.g., spatial GNN branch); evaluates collective module capacity rather than isolating the constraint.
Total Strict Isolation	**23**	**50.0%**	Only approximately half of the audited literature uses an ablation that attempts to isolate the physical constraint itself.

through the Elexon BMRS portal under the UK Open Government Licence v3.0. The complete code suite, AST auditing scripts, and the 46-study scoping extraction database with verbatim quotes and multi-coder consensus scores are openly accessible for peer review via the anonymous repository: https://github.com/anonymous-audit/pcwpf-audit.

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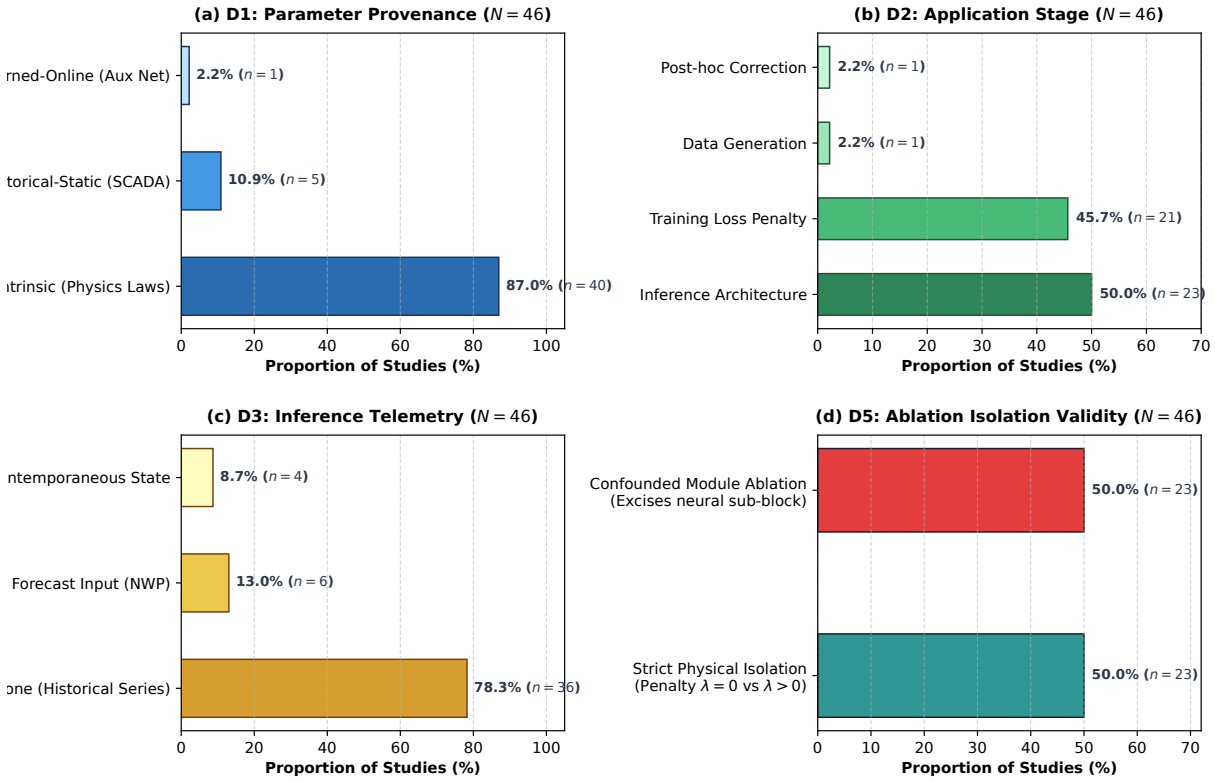


Figure B.8: Empirical Lifecycle Distributions across 46 Audited Studies ($N = 46$). (a) $\mathcal{D}_1$ Parameter Provenance; (b) $\mathcal{D}_2$ Application Stage; (c) $\mathcal{D}_3$ Inference Telemetry Requirements; (d) $\mathcal{D}_5$ Ablation Isolation Validity, showing the binary validity classification where Strict Physical Isolation ($n = 23$, 50.0%) aggregates 17 `constraint-on-off` and 6 `both` studies ($23 = 17 + 6$), contrasted against 23 Confounded Module Ablations (50.0%).

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